Automation and High-Skill Labor Markets: Evidence from Robo-Advisors

Abstract

Does artificial intelligence substitute or complement high-skill labor? We study financial robo-advisors and their impact on the labor market for human advisors. Using novel data on robo-advisors and advisor employment, we find that robo-advisors increase, rather than reduce, human advisor employment. To address endogenous adoption, we exploit cross-state regulatory entry costs interacted with a national financing shock affecting technology firms. States experiencing greater robo-advisor entry exhibit meaningful growth in advisor employment, particularly among firms serving retail investors. Investor-level evidence shows robo-advisor exposure increases subsequent engagement with human advisors, consistent with market expansion and complementarity between automation and professional financial services.

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Keywords: Artificial intelligence, robo-advisors, financial technology, high-skill labor, labor complementarity.

1 Introduction

Artificial intelligence is increasingly deployed in sectors traditionally staffed by highly trained professionals, including finance. The central economic question in such environments is not simply whether automation replaces labor, but whether these systems substitute for skilled workers at the margin or instead expand market scale and reallocate tasks toward domains where human input remains essential. Despite rapid diffusion of AI technologies, credible evidence on their effects in high-skill labor markets remains limited, largely because adoption is endogenous: firms introduce automation in response to shifting demand, regulatory conditions, and broader market trends. Identifying plausibly exogenous variation in AI entry is therefore critical for understanding how automation alters the demand for skilled labor.

This paper studies the labor-market consequences of AI-driven automation in the market for financial advice. The empirical setting is the diffusion of financial robo-advisors—algorithmic platforms that automate core components of portfolio management at low marginal cost. Robo-advisors mechanize standardized portfolio construction tasks historically performed by human advisors, while leaving complex financial planning, relationship management, and behavioral coaching to human professionals. This division of tasks yields theoretically ambiguous predictions. A substitution view predicts that robo-advisors displace human advisors by delivering a lower-cost alternative to traditional advisory services. A complementarity view predicts that automation lowers the cost of baseline advice, attracts new consumers into formal financial management, and expands demand for human advisors in segments where expertise and relational capital remain paramount (Autor, 2015; Acemoglu and Restrepo, 2018). In addition to expanding participation, automation may also increase advisor productivity by shifting routine portfolio construction toward algorithms while allowing human advisors to specialize in higher-value tasks. The central question is therefore whether artificial intelligence in a professional service industry substitutes for or complements high-skill labor.

To answer this question, we assemble a novel dataset linking a hand-collected census of robo-advisor providers and launch timing to comprehensive firm- and branch-level regulatory filings on advisor employment, branch locations, client composition, and firm characteristics. These data

allow us to track the number of financial advisors at the firm–state–year level over time, rather than relying on cross-sectional comparisons or platform-specific samples.

A key empirical challenge is that robo-advisor entry is endogenous to local financial conditions. Ordinary least squares estimates may confound robo growth with unobserved shocks that jointly affect technology startups and advisory employment. To address this concern, we implement an instrumental-variables strategy that exploits cross-state variation in minimum financial requirements for registered investment advisers relative to the NASAA model rule, interacted with the 2016 Protecting Americans from Tax Hikes (PATH) Act payroll tax credit provisions. States with more lenient financial requirements face lower fixed costs of registering and operating advisory firms, while the PATH Act disproportionately relaxed financing constraints for young, technology-oriented startups. The interaction of these features generates plausibly exogenous post-2016 variation in robo-advisor entry across states. In both state-level and firm–state–year specifications, the instrument strongly predicts robo diffusion, with first-stage F-statistics exceeding conventional thresholds.

Our central finding is that robo-advisors increase, rather than reduce, high-skill employment of human financial advisors. At the firm–state–year level, additional robo presence increases advisor employment within exposed firms by approximately 1.13 advisors per firm. Aggregated to the state level, a one-unit increase in the number of robo-advisors raises financial advisor employment by roughly 798 advisors, corresponding to about 5% of the average number of advisors in a state. Given the average level of robo presence in the post-2016 period, these estimates imply economically meaningful employment expansion and reject displacement effects of financial advisors.

We provide evidence that this complementarity operates primarily through market expansion rather than simple business reallocation, although competitive and productivity effects may also contribute. Employment responses are strongest among firms whose services overlap most closely with robo-advisors, particularly those serving retail clientele and offering wrap-fee or pension services, while effects are weaker among firms specializing in high-net-worth clients. This pattern is consistent with automation lowering the cost of baseline portfolio management while allowing human advisors to specialize in higher-value services.

Survey-based evidence further supports a market expansion mechanism. Exposure to robo-

advisors increases the probability that individuals engage with financial advisors, and individuals who adopt robo services are more likely to subsequently seek human advisory services. Additional demographic evidence shows that robo-advisor users tend to be younger investors who are newly entering the financial advice market. These individuals may not have participated in formal advisory services in the absence of robo platforms, suggesting that robo-advisors broaden participation in financial intermediation rather than simply reallocating existing clients across firms.

Consistent with workforce expansion, robo exposure both reduces instances of advisors leaving firms and increases hiring. Average advisor experience rises, and misconduct does not materially change, indicating no deterioration in workforce quality across observable dimensions. Finally, robo-advisory firms experience significant growth in assets under management and account counts following robo expansion. At the firm-branch level, robo-advisory firms experience increases in assets under management of approximately $864 million and 2,240 accounts in response to robo shocks, reinforcing the interpretation that algorithmic automation enlarges the overall scale of advisory intermediation.

This paper contributes to three strands of literature. First, it contributes to the literature on automation and artificial intelligence in labor markets. While prior work documents displacement effects of technology in routine and middle-skill occupations (Autor et al., 2003; Acemoglu and Autor, 2010; Bessen, 2016), more recent evidence highlights heterogeneity in how advanced technologies affect skilled workers (Webb, 2019; Grennan and Michaely, 2020; Jiang et al., 2021). We provide causal evidence from a high-skill professional service market in which AI adoption expands skilled employment. Our results underscore that the labor-market consequences of AI depend on whether automation replicates existing tasks or instead expands market scale and demand for complementary human inputs.

Second, the paper contributes to the literature on financial intermediation and technological entry. Fintech platforms expand in environments shaped by regulatory entry costs and financing conditions. By exploiting cross-state regulatory variation interacted with a national financing shock, we link regulatory design to technology diffusion and show how technology-induced entry can alter labor demand within financial intermediation through market expansion.

Third, we contribute to the robo-advisory and fintech literature, which has primarily examined

portfolio performance, investor behavior, and adoption patterns (D'Acunto et al., 2019; Rossi and Utkus, 2020; Back et al., 2021). We provide causal evidence on the labor-market equilibrium consequences of robo diffusion, addressing broader concerns about professional displacement in finance.

In sum, our evidence supports a complementarity view of AI in high-skill service industries. When automation lowers baseline service costs and expands participation, it can increase the equilibrium demand for skilled labor by reshaping task allocation and increasing the scale of professional services rather than replacing them.

The remainder of the paper proceeds as follows. Section 2 describes the institutional background of the financial advice industry and develops competing hypotheses of substitution and complementarity. Section 3 introduces the data and empirical strategy, including the regulatory instrument for robo-advisor entry. Section 5 presents the main employment results and examines mechanisms through market expansion and labor-market margins. The final section concludes.

2 Institutional Background and Economic Framework

Understanding how artificial intelligence affects high-skill labor markets requires attention to both the task structure of the occupation and the regulatory environment governing entry. The market for financial advice provides a setting in which both dimensions are observable. Robo-advisors automate core components of portfolio management within a highly regulated professional service industry, allowing us to study how AI entry affects the employment of licensed human advisors and the organization of advisory services.

2.1 Regulation and Entry in Investment Advice

Investment advisers operate within a state- and federally regulated framework. Firms must register as Registered Investment Advisers (RIAs) and file Form ADV disclosures detailing assets under management, client composition, compensation structures, and services offered. Individual advisors are also registered and tracked through regulatory filings that include employment history and misconduct disclosures. This structure is central to our analysis for two reasons. First, it provides

detailed and standardized measurement of high-skill labor supply, allowing us to observe advisor counts at the firm–state–year level. Second, it renders entry costs and regulatory constraints economically meaningful and observable.

States differ in minimum financial requirements imposed on registered investment advisers relative to the NASAA model rule. These requirements function as fixed entry costs: before operating in a state, advisory firms must satisfy state-level capital or bonding thresholds. The burden of these requirements is particularly salient for young or technology-oriented entrants with limited initial capital. This regulatory heterogeneity underpins our identification strategy. We exploit cross-state differences in these entry costs interacted with a national shock that relaxed financing constraints for young technology firms, generating plausibly exogenous variation in robo-advisor diffusion across states. Because the regulatory classification is time-invariant and does not mechanically determine employment levels at incumbent firms, identification arises from differential post-shock expansion of robo-advisors in lower-entry-cost environments.

In addition to regulatory requirements, Form ADV disclosures also report the types of advisory services firms provide and the composition of their client base. Several of these service categories are particularly relevant for understanding the interaction between robo-advisors and traditional advisory firms. Firms serving predominantly retail clientele typically provide standardized portfolio management to households with relatively modest account balances. Wrap-fee programs bundle portfolio management, trading, and advisory services into a single asset-based fee and often rely on model portfolios and automated rebalancing strategies. Similarly, pension and retirement advisory services frequently involve standardized asset allocation and lifecycle portfolio management for retirement accounts.

These services overlap closely with the core functions automated by robo-advisory platforms, which primarily provide algorithm-based portfolio construction and rebalancing at low cost. As a result, firms operating in these segments are more likely to experience changes in demand when robo-advisory technology diffuses. This overlap motivates our later heterogeneity analyses examining whether employment effects are strongest among firms offering services most closely aligned with robo-advisory capabilities.

2.2 Task Structure of Financial Advice and Robo Automation

The production of financial advice combines two broad categories of tasks. The first consists of scalable, technical portfolio-management activities such as asset allocation and rebalancing. These tasks are largely rules-based and therefore well suited to algorithmic automation. The second category consists of relationship-intensive and context-specific services, including holistic financial planning, retirement and estate planning, tax coordination, and behavioral coaching during periods of market stress. These activities rely on interpersonal trust, communication, and judgment.

Robo-advisors automate much of the first category by offering standardized portfolio management at low marginal cost. They do not replicate the full scope of advisory services but instead mechanize a subset of advisory production. The labor-market implications therefore depend on how automation affects equilibrium demand. If automated portfolio management is a close substitute for the marginal product of human advisors, robo entry should reduce advisor demand, particularly in standardized retail segments. If instead automation lowers the cost of baseline portfolio services and expands participation in formal wealth management, it may increase the scale of advisory activity and raise demand for complementary services provided by human advisors.

In addition to expanding market participation, automation may also affect the competitive environment within the advisory industry. By shifting routine portfolio management toward algorithms, robo-advisors may increase the productivity of human advisors and allow them to reallocate effort toward higher-value services such as financial planning and client relationship management. In this case, automation may increase the competitiveness of advisory firms while simultaneously increasing the value of human-intensive services. Additional descriptive evidence on the evolution of robo-advisory firms, their service offerings, and their client composition is provided in Appendix A1.

The diffusion of robo-advisory technology within the financial advisory industry follows a two-stage pattern. The earliest robo-advisors appear in the mid-2000s as fintech entrants offering automated portfolio management through digital platforms. Entry remains limited during the early years, but adoption accelerates after roughly 2013 as automated investment platforms become more widely available. Beginning around 2015, incumbent advisory firms increasingly introduce robo-advisory services alongside their traditional offerings, leading to the emergence of hybrid advisory models that combine automated portfolio management with human financial advice. This pattern

indicates that robo technology diffuses not only through entry by new firms but also through adoption by established advisory firms, an important feature of the industry when evaluating whether automation substitutes for or complements human advisors.

2.3 Economic Predictions

The labor-market effects of robo-advisor diffusion depend on whether automation primarily substitutes for existing advisory tasks or instead expands the scale and organization of advisory activity. In a market-expansion framework, lowering the cost of portfolio management increases participation among households that would otherwise remain outside formal advisory relationships. Greater participation raises aggregate demand for complementary services that remain human-intensive. Advisors may also reallocate effort toward higher-value tasks as routine portfolio management becomes automated.

Automation may also influence the competitive structure of the advisory market. By reducing the cost of standardized portfolio management and enabling scalable service delivery, robo-advisors can increase competitive pressure among firms while simultaneously increasing the productivity of human advisors who focus on differentiated services. In this case, automation reshapes the allocation of tasks within advisory firms while expanding the market for advisory services.

These mechanisms generate clear empirical predictions. Under substitution, greater exposure to robo entry should reduce advisor employment, particularly in standardized retail segments. Under complementarity through market expansion and productivity gains, robo presence should increase advisor employment, with stronger effects among firms serving retail clientele or offering services closely aligned with robo portfolios. The institutional features of the financial advice industry therefore provide both the variation necessary for identification and a framework for evaluating whether AI-driven automation substitutes for or complements high-skill labor.

3 Data

This section describes the data sources used to measure advisor employment, robo-advisor entry, investor demand for advice, and state-level controls. The full sample period spans 2006 through

2019. The start date corresponds to the emergence of robo-advisory providers in our data, and the end date precedes the structural disruptions associated with the COVID-19 pandemic.

3.1 Advisor Employment and Firm Characteristics

The primary source for firm-level information is Form ADV Part I, filed by Registered Investment Advisers (RIAs) with the Securities and Exchange Commission (SEC). Advisory firms must file Form ADV annually and whenever material changes occur, generating repeated observations within a year. The filings report states of registration, the number of financial advisors employed, assets under management (AUM), number of client accounts, and client composition.

All registered financial advisors file Form U4 through FINRA, which records employment history, state registrations, and misconduct disclosures. We obtain advisor-level data from FINRA BrokerCheck and complement it with records from the SEC's Investment Adviser Public Disclosure (IAPD) system. These data allow us to construct the universe of individuals practicing as brokers, financial advisors, or both, and to map employment to firm–state–year observations.

To extend the advisor employment series prior to 2012, we aggregate individual advisor records to the firm–state–year level and infer the number of advisors employed by each firm in each state. We validate this backfilling procedure by comparing aggregated counts to the employment ranges reported in Form ADV prior to 2012; the constructed counts fall within reported ranges in approximately 95% of cases. The resulting panel is structured at the firm–state–year level and forms the primary dataset for our employment analyses.

3.2 Robo-Advisor Entry

We construct a hand-collected census of robo-advisory service providers and their launch dates. Because no centralized database exists, robo-advisors are identified using industry news sources, trade publications, firm websites, archived press releases, and public records. When launch timing is not directly reported, we corroborate dates using archived digital sources and regulatory filings.

The final sample identifies 174 firms offering robo-advisory services between 2006 and 2019. Of these, 98 firms (56.3%) entered as dedicated robo-advisory startups, while 36 firms (20.7%)

are incumbent advisory firms that subsequently adopted robo platforms, forming hybrid business models. The remaining providers include 27 business-to-business platforms (15.5%) offering robo infrastructure to other RIAs and 13 exempt reporting advisers (7.5%) primarily serving private or venture capital funds. In total, 134 firms (77.0%) provide robo-advisory services directly to retail or individual investors. Consistent with a scalable advisory model, robo startups disproportionately offer wrap-fee programs and serve predominantly retail, non–high-net-worth clientele.

For the empirical analysis, robo exposure is measured as the number of robo-advisory providers operating in a given state–year. Both dedicated startups and incumbent adopters contribute to state-level robo presence, as each expands the availability of algorithmic portfolio management within that state. Figure 1 illustrates the geographic diffusion of robo-advisory activity over the sample period. Robo presence expands from a small set of financial and technology centers in the mid-2000s to broad national coverage by 2019, with substantial cross-state variation in intensity.

3.3 Investor Demand for Financial Advice

To examine demand-side responses, we use data from FINRA's National Financial Capability Study (NFCS), conducted triennially beginning in 2012. The survey records whether respondents use a financial advisor and, in later waves, whether they use robo-advisory services, along with demographic characteristics and state of residence.

These data allow us to construct state-level measures of advisor usage and to estimate individual-level regressions linking robo exposure to the probability of using a financial advisor, conditional on demographic and financial controls. The NFCS therefore provides a complementary demand-side perspective to the firm-level employment analysis.

3.4 State-Level Controls and Auxiliary Data

We incorporate state-level macroeconomic controls from the Bureau of Economic Analysis, Bureau of Labor Statistics, and the U.S. Census Bureau. These include financial-sector employment, unemployment rates, income, GDP, and demographic composition. In robustness exercises, we incorporate external measures of fintech activity, including venture deal counts from PitchBook, to

capture broader technology-sector dynamics.

3.5 Descriptive Statistics

Table 1 reports summary statistics for the main variables used in the analysis, and Table A3 provides detailed variable definitions. Panel A presents firm–state–year measures describing the size and composition of advisory firms. The typical firm in our sample is small: the median firm employs one financial advisor and reports few brokers or other employees, although the distribution is highly skewed. Average advisor experience is approximately 14 years, and hiring and separation flows are modest relative to firm size. These patterns reflect the structure of the financial advisory industry, which consists largely of small advisory practices alongside a small number of very large national firms.

Panel B reports state–year variables used in the instrumental-variable design. The average state-year contains approximately six robo-advisory firms, with substantial growth over time as reflected in the positive change in the number of robo-advisors. Slightly more than half of the states in the sample are classified as having relatively lenient regulatory entry requirements, and approximately 42% of observations occur in the post-2016 period. The interaction of these components forms the instrumental variable used to generate differential robo-advisor diffusion across states.

Panel C reports firm-level measures of discretionary assets under management and discretionary accounts, which capture the scale of advisory activity. Panel D reports investor-level measures of advisory service usage. Consistent with the relatively recent emergence of robo-advisors, the probability that an investor uses a robo-advisory service remains low on average, while the use of traditional financial advisors is substantially more common. Additional descriptive statistics on advisory firm characteristics, client composition, and the evolution of robo-advisory firms are reported in Appendix A1.4.

While the data cover the full 2006–2019 period, our primary analysis focuses on the post-2012 sample. Prior to 2012, Form ADV filings frequently reported employment counts for financial advisors and brokers in ranges rather than exact numbers. Following the 2011 revision to Form ADV, firms were required to report precise employment counts beginning in 2012, allowing for more accurate measurement of advisor employment. In addition, the identification strategy exploits

differential expansion of robo-advisors following the 2016 PATH Act financing shock. Using the 2012–2016 period as a pre-treatment window allows us to observe advisor employment trends prior to the policy shock and evaluate parallel trends across states. For these reasons, the IV specifications are estimated over the 2012–2019 period. Appendix A2 shows that the results are qualitatively similar when the analysis is extended to the full 2006–2019 sample using alternative specifications.

4 Empirical Strategy and Identification

This section describes our empirical strategy for estimating the causal effect of robo-advisor entry on high-skill employment in the financial advice industry. A central challenge is that robo-advisor entry is endogenous to local financial conditions, which may bias ordinary least squares (OLS) estimates. To address this concern, we implement an instrumental-variables strategy that exploits cross-state differences in regulatory entry costs for investment advisers interacted with a national shock to financing conditions for young technology firms.

4.1 Baseline Specifications

Let $R_{s,t}$ denote the number of robo-advisory service providers operating in state s in year t, and let $FA_{i,s,t}$ denote the number of human financial advisors employed by firm i in state s and year t. Our primary firm–state–year specification is:

$$FA_{i,s,t} = \beta R_{s,t} + \Gamma' X_{i,s,t} + \Lambda' Z_{s,t} + \alpha_i + \alpha_s + \alpha_t + \alpha_{i\times t} + \varepsilon_{i,s,t}, \tag{1}$$

where $X_{i,s,t}$ includes firm-by-state-by-time controls (e.g., broker employment), $Z_{s,t}$ includes state-by-time controls (e.g., number of employees in the financial services sector, share of college-educated population, GDP, unemployment, etc.), α_i are firm fixed effects, α_s are state fixed effects, α_t are year fixed effects, and $\alpha_{i\times t}$ are firm-by-year fixed effects that absorb time-varying firm-level shocks common across the firm's geographies. Standard errors are clustered at the firm level.

We also estimate state-level analogs that aggregate employment across firms. Let $FA_{s,t}$ denote total financial advisor employment in state s and year t. The state-year specification is:

$$FA_{s,t} = \beta R_{s,t} + \Lambda' Z_{s,t} + \alpha_s + \alpha_t + u_{s,t}, \tag{2}$$

where $Z_{s,t}$ includes state-level time-varying controls, and standard errors are clustered at the state level.

Across specifications, the coefficient of interest β captures how robo-advisor presence affects the equilibrium level of human advisor employment. A positive coefficient would be consistent with complementarity between robo-advisors and human advisors, which may arise either through market expansion, whereby robo platforms attract new investors into advisory markets, or through competitive responses by incumbent firms that increase hiring as robo technology diffuses.

4.2 Why OLS Is Biased

The regressor of interest, $R_{s,t}$, is an equilibrium outcome. Robo-advisors may enter states where demand for low-cost advice is rising, where local conditions favor fintech startups, or where incumbent advisory firms are already expanding. Each of these factors may independently affect advisor employment. The direction of bias is therefore ambiguous: if robo entry is strongest in states where incumbent advisors face competitive pressure, OLS may understate the causal effect; if robo entry coincides with unobserved increases in retail participation and demand for advisory services, OLS may overstate it. We therefore rely on an instrumental-variables strategy that isolates variation in robo presence driven by regulatory entry costs and an external financing shock.

4.3 Instrument Construction: Regulation $\times$ Financing Shock

Our identification exploits a difference-in-differences-style interaction between (i) cross-state differences in regulatory entry costs for investment advisers and (ii) a post-2016 national shock that relaxed financing constraints for young technology firms.

The identifying variation arises from differential post-2016 expansion of robo-advisors in states with lower regulatory entry costs. Because regulatory stringency is fixed over time and the financing shock is national, the interaction captures whether robo entry responds more strongly in states where fixed costs of advisory registration are lower.

Cross-sectional variation: state minimum financial requirements. States impose minimum net worth or bonding requirements on registered investment advisers, following NASAA Model Rule 202(d)-1.[1] While NASAA provides model guidance, states adopt these requirements with varying degrees of stringency. These thresholds represent fixed entry costs: higher requirements increase the capital that firms must maintain to operate.

We classify a state as $Lenient_s = 1$ if its minimum financial requirements are below NASAA recommendations, and zero otherwise. Because robo-advisors must register as investment advisers, these regulatory differences directly translate into cross-state variation in entry costs. The regulatory classification is time-invariant over our sample. Figure 2 illustrates the geographic distribution of lenient and stringent states based on this classification.

Time-series variation: post-2016 financing shock. Beginning in 2016, the Protecting Americans from Tax Hikes (PATH) Act introduced financing provisions that relaxed constraints for young technology firms, allowing eligible startups to apply research and development (R&D) tax credits against payroll tax liabilities. This provision enabled firms without taxable income to monetize R&D tax credits during early stages of development, effectively reducing financing frictions for venture-backed technology startups, including firms developing digital financial services.

Because robo-advisors are technology-based entrants that must also satisfy investment-advisor registration requirements, the relaxation of financing constraints disproportionately facilitated robo-advisor expansion, particularly in states where regulatory entry costs were lower.

Evidence from the startup ecosystem is consistent with such a financing shock. Figure 3 shows that both venture capital deal activity and the number of FinTech startups increase substantially during the post-2016 period. These patterns are consistent with prior research documenting that the PATH Act's R&D tax credit provisions increased innovation and startup activity among technology firms (Chen and Hsieh, 2024).

Figure 4 illustrates the resulting pattern for robo-advisors: prior to 2016, robo presence evolves similarly across lenient and stringent states; after 2016, robo expansion is concentrated in lenient states.

[1] See NASAA Model Rule 202(d)-1 (as amended in 2011), which specifies minimum net worth or bonding requirements depending on adviser activity.

Instrument. We construct the instrument:

$$IV_{s,t} = Lenient_s \times Post2016_t, \tag{3}$$

where $Post2016_t$ equals one for $t \geq 2016$. The instrument captures the differential post-2016 increase in robo-advisor presence in lower-entry-cost states.

4.4 Two-Stage Least Squares

We estimate equations (1) and (2) using two-stage least squares (2SLS). The firm-level first stage is:

$$R_{s,t} = \pi IV_{s,t} + \Theta' X_{i,s,t} + \Gamma' Z_{s,t} + \alpha_i + \alpha_s + \alpha_t + \alpha_{i\times t} + \nu_{i,s,t}, \tag{4}$$

and the second stage replaces $R_{s,t}$ with its fitted value $\widehat{R}_{s,t}$ in equation (1). The state-level 2SLS specification analogously uses $Z_{s,t}$ and state and year fixed effects.

4.5 First-Stage Evidence and Identification

The interaction between lenient state regulation and the post-2016 period strongly predicts robo-advisor diffusion. In the state-level specification, the first-stage F-statistic equals 10.48, exceeding conventional weak-instrument thresholds. In the firm–state–year specification, the first stage is substantially stronger (F-statistic above 2,000), reflecting the larger sample and the sharp post-2016 divergence documented in Figure 4.

The exclusion restriction requires that $IV_{s,t}$ affects advisor employment only through robo-advisor entry. A primary concern is that lenient states may have experienced distinct post-2016 economic or technological expansions unrelated to robo diffusion. We therefore examine whether identification is driven by post-2016 divergence rather than pre-existing differential trends.

We estimate a dynamic specification interacting the lenient-state indicator with year indicators:

$$R_{s,t} = \sum_k \beta_k \left(Lenient_s \times \mathbb{1}\{t = k\} \right) + X'_{s,t}\gamma + \alpha_s + \alpha_t + \varepsilon_{s,t}. \tag{5}$$

Figure 5 shows no differential pre-trends between lenient and stringent states prior to 2016, while the divergence emerges only after the financing shock.

Figure 6 provides complementary evidence by plotting the difference in the probability that a robo-advisor begins operating in treatment states relative to control states. The probability of robo-advisor entry evolves similarly across state groups prior to 2016 but increases noticeably in lenient states after the financing shock.

To illustrate how this differential expansion translates into advisor employment, Figure 7 plots the difference in advisor employment between lenient and stringent states over time. Consistent with the first-stage evidence, there is no differential trend in advisor employment prior to 2016. Following the post-2016 expansion of robo-advisors, advisor employment increases in lenient states relative to stringent states. This reduced-form pattern is consistent with the IV design.

As an additional timing test, we implement a placebo assuming the shock occurred in 2014. Re-estimating the first stage using 2011–2013 as the pre-period and 2014–2016 as the placebo post-period yields a small and statistically insignificant coefficient.

Several additional features support the exclusion restriction. The regulatory component $Lenient_s$ is time-invariant, so identification arises from differential post-2016 expansion rather than contemporaneous regulatory changes. The financing shock primarily affected eligible young technology firms, limiting direct effects on incumbent advisory employment.

Moreover, Figure 8 plots the average number of traditional financial advisory firms operating in lenient and stringent states over time. Unlike robo-advisor entry, the number of advisory firms evolves smoothly across both groups with no visible divergence following the 2016 financing shock. This pattern indicates that the instrument does not capture a broader expansion of the financial advisory industry in lenient states, but rather differential diffusion of robo-advisory technology. Consistent with this evidence, regressions of advisory-firm counts on $IV_{s,t}$ yield economically small and statistically insignificant estimates. Together, these results support the interpretation that the instrument isolates exogenous variation in robo-advisor diffusion.

5 Main Results

Table 2 reports baseline estimates of the effect of robo-advisor presence on financial advisor employment at both the state-year and firm–state–year levels.

Column (1) of Panel C presents the 2SLS estimate from the state-year specification. The coefficient implies that the entry of one additional robo-advisor increases total financial advisor employment in a state by approximately 798 advisors ($p < 0.05$), or roughly 5% of the average advisor count. While this aggregate effect appears sizable, it reflects the scalable nature of robo platforms, which can expand their client base without meaningful capacity constraints. In this setting, robo-advisors may increase overall demand for advisory services by attracting new investors into formal portfolio management, thereby expanding the market for financial advice.

Consistent with this interpretation, the firm–state–year estimate in Column (2) indicates that a one-unit increase in state-level robo presence raises advisor employment within a given firm-state by 1.13 advisors on average ($p < 0.05$).[2] Thus, at the micro level, the effect corresponds to roughly one additional advisor per firm per robo entrant. These aggregate effects are consistent with a complementarity between robo-advisors and human advisors; subsequent analyses examine the mechanisms underlying this relationship.

Panel A reports the corresponding OLS estimates. Although the OLS coefficients are positive, they are substantially smaller in magnitude (472 at the state level and 0.18 at the firm–state level). The larger IV estimates suggest downward bias in OLS, consistent with robo entry occurring in states where advisor employment would otherwise have grown more slowly. One interpretation is that robo-advisors initially enter markets where traditional advisory services face competitive pressure or slower growth, which attenuates the simple OLS relationship between robo presence and advisor employment. We do not pursue a structural interpretation of this bias but note that it reinforces the importance of addressing endogenous technology adoption.

Table 3 examines the sensitivity of the firm–state results to additional controls. Column (1) reproduces the baseline 2SLS specification. Column (2) adds broker employment at the firm–

[2]The distribution of advisor counts is highly skewed due to the presence of very large advisory firms. Re-estimating the baseline specification after excluding extreme observations of advisor employment yields a similar estimate: the coefficient on $\widehat{R}_{s,t}$ is 0.74 (SE = 0.37), statistically significant at the 5% level.

state level, allowing for potential substitution or complementarity between advisory and brokerage functions. Column (3) further includes firm-level scale measures—log discretionary assets under management and log number of accounts—to account for expansion in client base or overall firm growth.

Across specifications, the estimated effect of robo exposure remains stable, ranging from 1.01 to 1.10 additional advisors per firm–state. Controlling for broker employment does not attenuate the coefficient, indicating that the result is not driven by internal reclassification between brokers and advisors. Similarly, conditioning on assets under management and account counts does not materially alter the estimate. While these scale variables are strongly correlated with advisor employment, the persistence of the robo coefficient suggests that robo-advisor diffusion is associated with an employment response beyond general firm expansion. This pattern is consistent with robo platforms expanding the overall advisory market while also inducing incumbent firms to increase hiring as robo technology diffuses.

As an additional robustness check, we implement an alternative Bartik-style instrument based on historical internet advisory activity and global FinTech investment growth. The construction, identifying assumptions, and results for this specification are presented in Appendix A2. The estimates from this alternative design are similar in magnitude and statistical significance to the baseline results.

6 Mechanisms: Market Expansion and Complementarity

The positive employment effects documented above are inconsistent with a pure automation–substitution framework. Instead, the evidence points to a complementary mechanism in which robo-advisory diffusion expands the overall market for advisory services and increases demand for human advisory labor. In particular, robo platforms may increase participation in advisory markets by lowering the cost of entry into professional portfolio management, while incumbent advisory firms respond to this expansion by increasing hiring to meet demand for complementary human-intensive services.

Robo-advisors automate standardized portfolio-management tasks, reducing marginal costs and lowering minimum account thresholds. By expanding access to advisory services for retail investors, robo platforms increase participation in formal wealth management. In a setting where human advisors provide complementary services—such as financial planning, tax coordination, and behavioral guidance—an expansion in market participation raises demand for these human-intensive tasks.

Figure 9 provides descriptive evidence consistent with this mechanism. Panel A plots the evolution of average discretionary assets under management (AuM), while Panel B reports the average number of discretionary accounts. Both measures increase substantially over the sample period, with particularly strong growth among firms offering robo-advisory services during the post-2016 diffusion period. The joint rise in assets and account counts suggests that the growth in advisory activity reflects broader participation rather than simply higher balances per client.

Table 4 provides evidence supporting these patterns. The estimates relate robo-advisor diffusion to discretionary assets under management (AuM) and the number of discretionary accounts managed by robo-advisory firms. Panel A reports state-level results. Robo diffusion is associated with economically large increases in advisory activity among robo-advisory firms. A one-unit increase in the number of robo-advisors in a state is associated with an additional $50.1 billion in discretionary assets under management and roughly 11.9 thousand discretionary accounts. The joint increase in assets and account counts suggests that robo diffusion corresponds with greater investment activity and broader participation in advisory services.

Panel B examines the same outcomes at the firm level and allows the effect to differ for firms offering robo-advisory services. The baseline coefficient on $R_{s,t}$ is small and statistically insignificant. In contrast, the coefficient on the interaction between robo diffusion and the indicator for robo-advisory firms implies that robo-advisory firms experience an additional $864 million in discretionary assets under management and approximately 2.24 thousand additional accounts for each unit increase in robo diffusion. These results indicate that robo diffusion is associated with a substantial expansion in the scale of the advisory industry, both in terms of assets managed and the number of participating investors. The pattern suggests that robo technology lowers barriers to entry into professional wealth management, thereby expanding participation in advisory markets. Consistent with this interpretation, Appendix A3 presents descriptive and regression evi-

dence showing that robo-advisor users tend to be younger and more employment-active households than traditional advisor clients. These demographic patterns suggest that robo platforms attract investors who may not have otherwise entered the financial advisory market under traditional account minimums and fee structures.

6.1 Scope of the Employment Effect

The baseline results establish that robo diffusion increases advisory employment. A natural question is whether this reflects general firm expansion or a targeted increase in advisory labor.

Table 5 addresses this distinction. Column (1) restricts the sample to firms providing advisory services. The 2SLS estimate implies that one additional robo-advisor increases advisory employment at such firms by 4.06 advisors (standard error 1.38), a magnitude larger than the average firm-level effect in the baseline specification. This concentration suggests that the employment response is directly linked to advisory production rather than broad firm growth.

Columns (2) and (3) further clarify the scope. Column (2) examines brokerage employment and finds no statistically significant response. Column (3), which uses total high-skilled employment (advisors plus brokers), likewise yields no significant effect. The absence of a brokerage response indicates that robo diffusion does not increase high-skilled headcount indiscriminately; instead, the effect is specific to advisory labor, consistent with task-level complementarity between automated portfolio management and human advisory services.

6.2 Investor-Level Evidence on Advisory Demand

If robo-advisory platforms expand participation in advisory markets, states with greater robo diffusion should exhibit higher usage of financial advisors. Moreover, robo adoption at the individual level should not crowd out human advisory usage if services are complementary.

Table 6 provides evidence consistent with these predictions. Column (1) reports 2SLS estimates linking state-level robo diffusion to the probability that an individual uses a financial advisor. The estimated coefficient implies a statistically significant increase in advisor usage associated with robo entry (standard error 0.09). Column (2) examines individual-level behavior: investors

who report using robo-advisory services are 6 percentage points more likely to also use a human advisor (standard error 0.02). These patterns indicate that robo platforms increase engagement with financial advice rather than displacing traditional advisory relationships.

6.3 Service Similarity and Complementary Demand

If robo technology expands demand in service domains closely aligned with standardized portfolio management, employment effects should be strongest among firms operating in adjacent segments. Firms serving retail clientele, offering wrap-fee programs, or providing pension services operate in areas most similar to robo-advisory offerings.

Table 7 evaluates this prediction. Across specifications, the baseline effect of robo diffusion remains positive. The interaction terms show that the employment response is significantly stronger among firms whose service models overlap with robo platforms. In Column (1), firms serving predominantly retail clientele experience an additional 1.13 advisor increase per unit of robo diffusion (standard error 0.24). Column (2) reports an incremental increase of 0.82 advisors (standard error 0.28) for firms offering wrap-fee programs, and Column (3) reports an incremental increase of 0.60 advisors (standard error 0.24) for firms providing pension services. The concentration of effects in adjacent service segments supports a complementary production structure rather than a general equilibrium reallocation across unrelated areas of finance.

6.4 Retail vs. High-Net-Worth Segments

A sharper implication of the market-expansion mechanism is that employment effects should be concentrated in retail-oriented firms, while firms serving predominantly high-net-worth (HNW) clients—whose services are less standardized and less directly targeted by robo platforms—should exhibit weaker responses.

Table 8 supports this prediction. Column (1) shows that firms serving primarily retail clientele experience an additional 0.82 advisor increase (standard error 0.27) beyond the baseline effect. In contrast, Column (2) reports no statistically significant differential response for firms focused on HNW clientele. The absence of an HNW response reinforces the interpretation that robo diffusion

expands participation among retail investors rather than shifting activity within bespoke advisory segments.

6.5 Organizational Complementarity and Robo Adoption

Finally, we examine how firms organizationally respond to robo entry. Table 9 interacts robo diffusion with an indicator for firms that adopt robo-advisory services. The interaction coefficient implies an additional increase of 22.11 financial advisors (standard error 10.35) among robo-adopting firms beyond the baseline effect. This magnitude suggests that hybrid firms expand advisory capacity following robo integration.

The result is consistent with organizational complementarity: incorporating automated portfolio management may increase client acquisition, account volume, and cross-selling opportunities, all of which raise demand for human advisory services. Rather than replacing advisors internally, robo adoption appears to induce expansion in advisory labor within adopting firms.

7 Workforce Dynamics and Composition

7.1 Hiring and Separation

The preceding analysis documents an increase in advisory employment associated with robo diffusion, consistent with a market expansion mechanism. We next examine the labor-market margins through which this adjustment occurs. Changes in headcount can arise from reduced separations (advisors leaving their firms), increased hiring, or both. Distinguishing between these channels helps clarify whether robo exposure primarily stabilizes incumbent advisors or induces active workforce expansion.

Table 10 decomposes employment changes into separation and hiring flows at the firm–state level. Separations capture advisors who leave the firm in a given year, while hiring captures new advisor inflows. Column (1) shows that robo diffusion is associated with a statistically significant decline in separations. Column (2) reports a statistically significant increase in hiring. The joint pattern indicates that robo exposure both improves retention and increases recruitment, consistent

with net workforce expansion rather than reclassification or internal reshuffling.

7.2 Advisor Experience and Misconduct

Changes in workforce size may also affect workforce composition. Rapid expansion could, in principle, reduce workforce quality if firms scale by hiring less-experienced advisors or relaxing screening standards. Alternatively, if robo integration raises demand for higher-value advisory tasks, firms may retain and attract more experienced personnel.

Table 11 examines these outcomes. Column (1) shows that robo diffusion is associated with a statistically significant increase in average advisor experience at the firm–state level. This result suggests that employment growth does not come at the expense of human capital depth. Column (2) reports no statistically significant change in misconduct incidence. The estimated coefficient is economically small and indistinguishable from zero. Together, these findings indicate that workforce expansion associated with robo diffusion is not accompanied by observable deterioration in advisor quality as measured by experience or misconduct.

8 Conclusion

This paper examines whether artificial intelligence substitutes for or complements high-skill labor in a professional service industry. Using the diffusion of financial robo-advisors as an empirical setting, we provide causal evidence that AI expands, rather than displaces, human financial advisor employment.

To address endogenous technology adoption, we exploit cross-state differences in regulatory entry costs for investment advisers interacted with a national financing shock that disproportionately benefited young technology firms. This interaction generates plausibly exogenous variation in robo-advisor diffusion across states. Two-stage least squares estimates show that robo-advisor entry significantly increases human advisor employment at both the firm–state and state–year levels.

We further show that the employment response operates primarily through market expansion rather than simple reallocation. Robo diffusion increases hiring and reduces separations, raises average advisor experience, and does not increase misconduct incidence. Employment gains are

concentrated among firms serving retail clientele and offering services closely aligned with robo-advisory models, while firms focused on high-net-worth clients exhibit no differential response. Investor-level evidence indicates that robo exposure increases the probability of using financial advisory services and that robo users are more likely to engage human advisors, consistent with complementary demand.

These findings contribute to the literature on artificial intelligence and labor markets by demonstrating that the employment effects of automation in high-skill occupations depend critically on task structure and demand expansion. In the market for financial advice, robo-advisors automate scalable portfolio-management tasks while leaving relationship-intensive and complex planning activities to human professionals. By lowering the cost of baseline advisory services and reducing minimum participation thresholds, robo platforms broaden participation in formal wealth management and increase demand for complementary human inputs.

More broadly, our results highlight that the labor-market consequences of AI in professional services are shaped not only by technological capability but also by institutional and market conditions. When automation reduces the cost of standardized tasks and expands access to services that were previously constrained by scale or price, equilibrium employment effects may be positive even in occupations directly exposed to algorithmic automation. In this environment, automation reshapes the allocation of tasks within firms and increases the value of human-intensive services rather than eliminating them.

Understanding how AI affects high-skill labor therefore requires attention not only to task automation but also to market expansion, institutional context, and organizational adaptation. As AI systems continue to diffuse across professional service industries, the balance between substitution and complementarity will likely depend on whether these technologies simply replicate existing tasks or instead expand the scale and scope of service provision.

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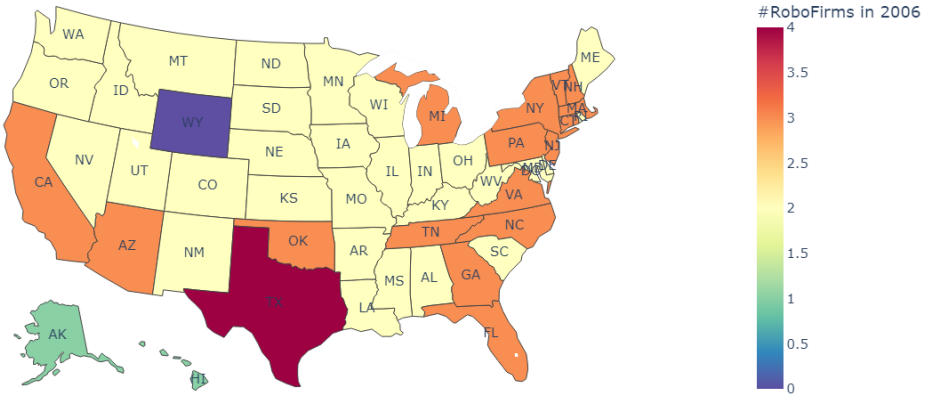
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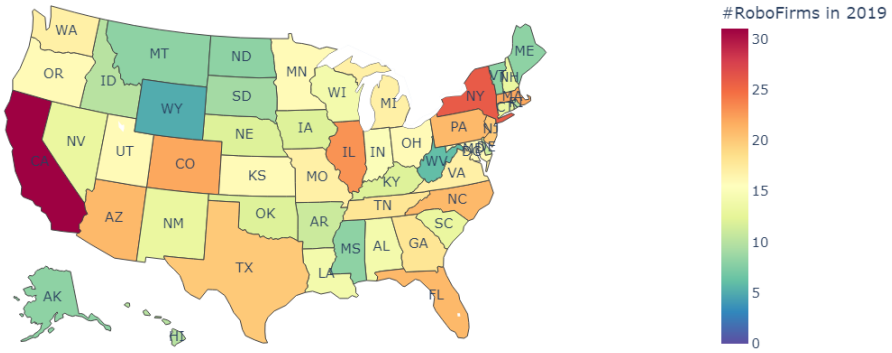
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(a) Distribution of Robo-Advisory Firms in 2006



(b) Distribution of Robo-Advisory Firms in 2019

Figure 1: **Geographic Expansion of Robo-Advisory Firms.**
This figure illustrates the number of robo-advisory service providers operating in each U.S. state in 2006 (Panel A) and 2019 (Panel B). Robo-advisory firms are identified through hand-collected data on launch dates and firm operations. The color scale reflects the number of robo-advisory providers active in a given state-year.

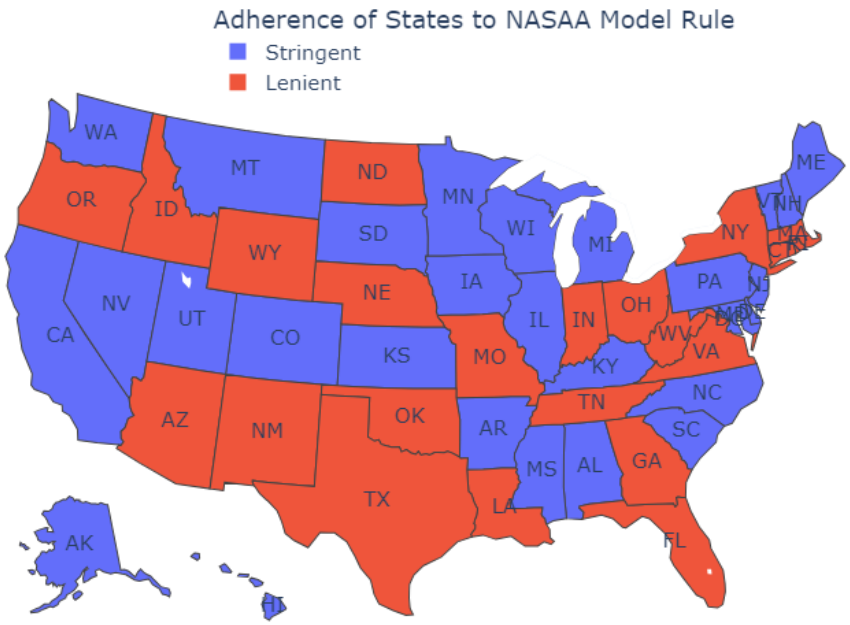
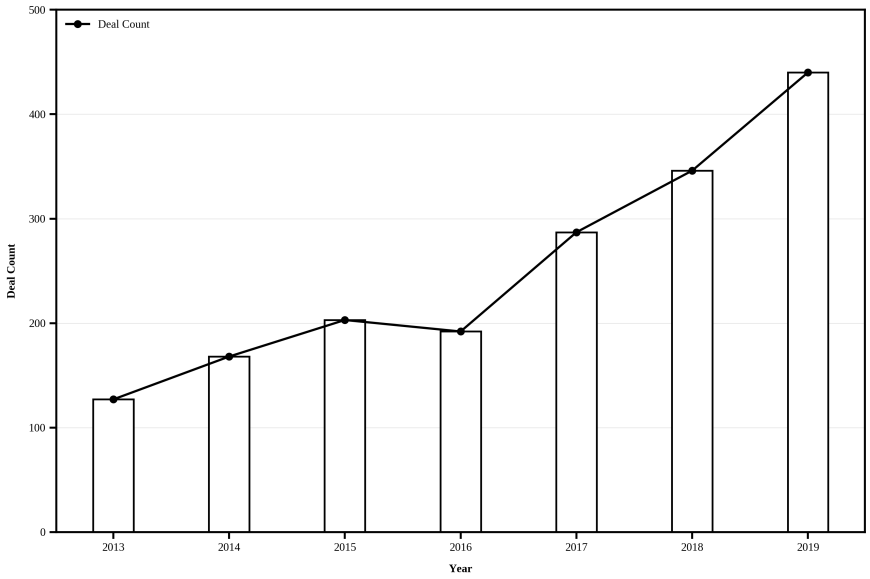
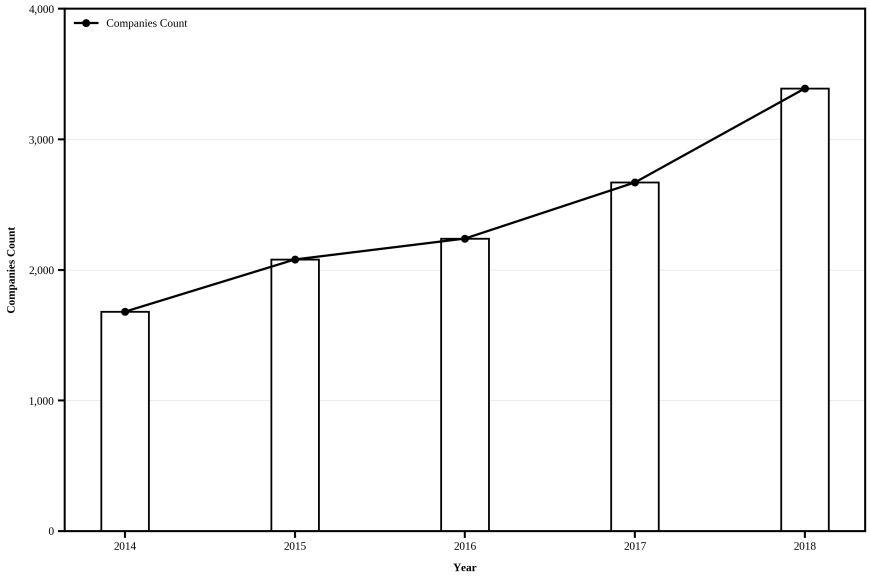


Figure 2: **State Classification by NASAA Model Rule Financial Requirements.** The map classifies U.S. states according to whether their minimum financial requirements for registered investment advisers are lenient (below NASAA recommendations) or stringent (on par with or stricter than NASAA recommendations).



(a) FinTech Venture Capital Deal Counts



(b) FinTech Startup Counts

Figure 3: **FinTech Investment and Startup Formation.** Panel (a) plots the annual number of venture capital deals involving FinTech firms, while Panel (b) plots the number of FinTech startup companies. Both series exhibit substantial growth following the 2016 financing policy change. Source: PitchBook.

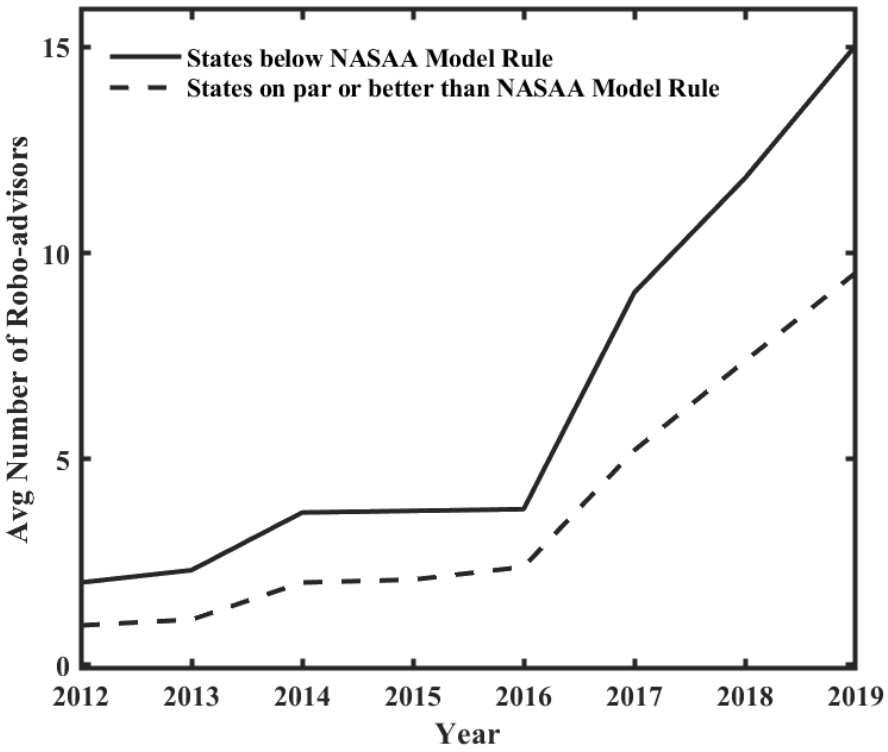


Figure 4: **Average Number of Robo-Advisors by Regulatory Stringency.**
The figure plots the average number of robo-advisory firms operating in states classified as lenient (below NASAA Model Rule financial requirements) and stringent (on par with or stricter than NASAA recommendations) from 2012 to 2019. Prior to 2016, robo presence trends are similar across groups. Following the 2016 PATH Act, robo-advisor presence increases more sharply in lenient states relative to stringent states.

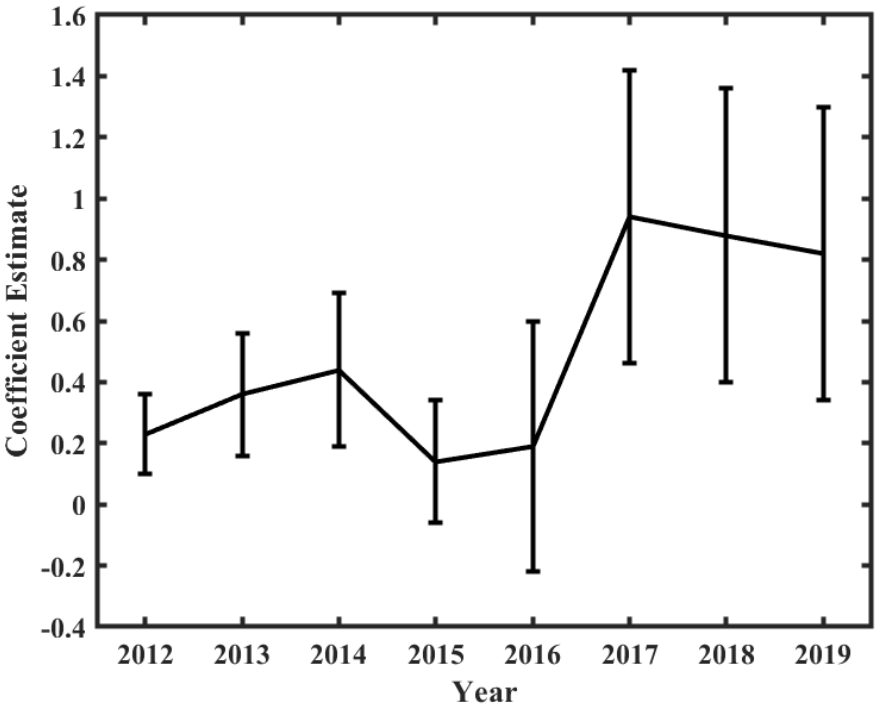


Figure 5: **Dynamic First-Stage Estimates of Robo-Advisor Entry.**
The figure plots the event-time coefficients β_t, which regress the number of robo-advisors $R_{s,t}$ on interactions between the lenient-state indicator and year indicators, controlling for state and year fixed effects and state-year covariates. Error bars denote 95% confidence intervals. The omitted category corresponds to the pre-2016 baseline period. The coefficients indicate no differential pre-trends between lenient and stringent states prior to 2016, while the increase in robo-advisor entry is concentrated in the post-2016 period.

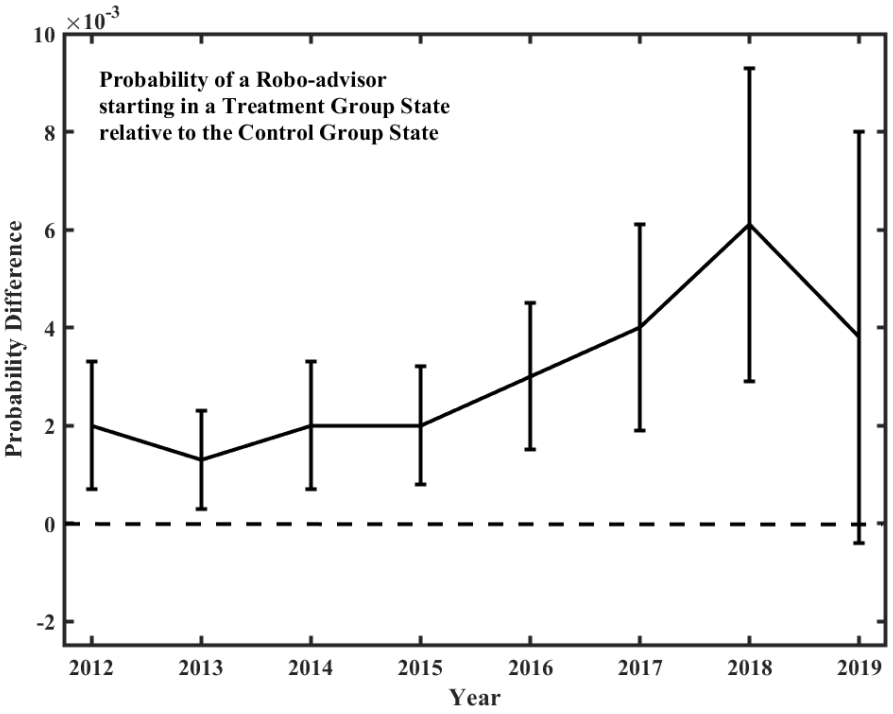


Figure 6: **Probability of Robo-Advisor Entry in Treatment vs. Control States.** The figure plots the difference in the probability that a robo-advisor begins operating in treatment states relative to control states over time. Error bars represent 95% confidence intervals. The dashed horizontal line denotes zero. The figure illustrates the absence of differential trends prior to 2016 and the subsequent increase in robo-advisor entry in treatment states following the financing shock.

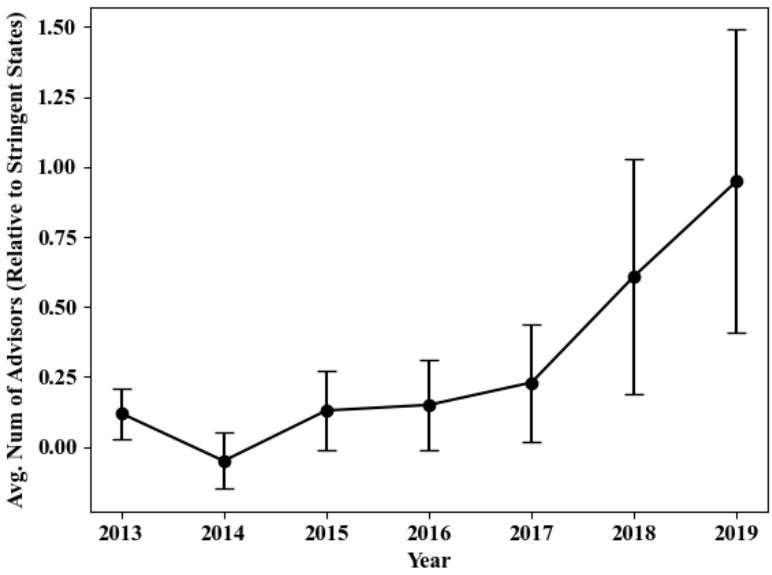


Figure 7: **Advisor Employment in Lenient vs. Stringent States.** The figure plots the average number of financial advisors in states classified as lenient relative to stringent states over time. Error bars denote 95% confidence intervals. The coefficients are normalized relative to the pre-2016 period. The figure shows no differential pre-trends prior to 2016, while advisor employment increases in lenient states following the expansion of robo-advisors after 2016.

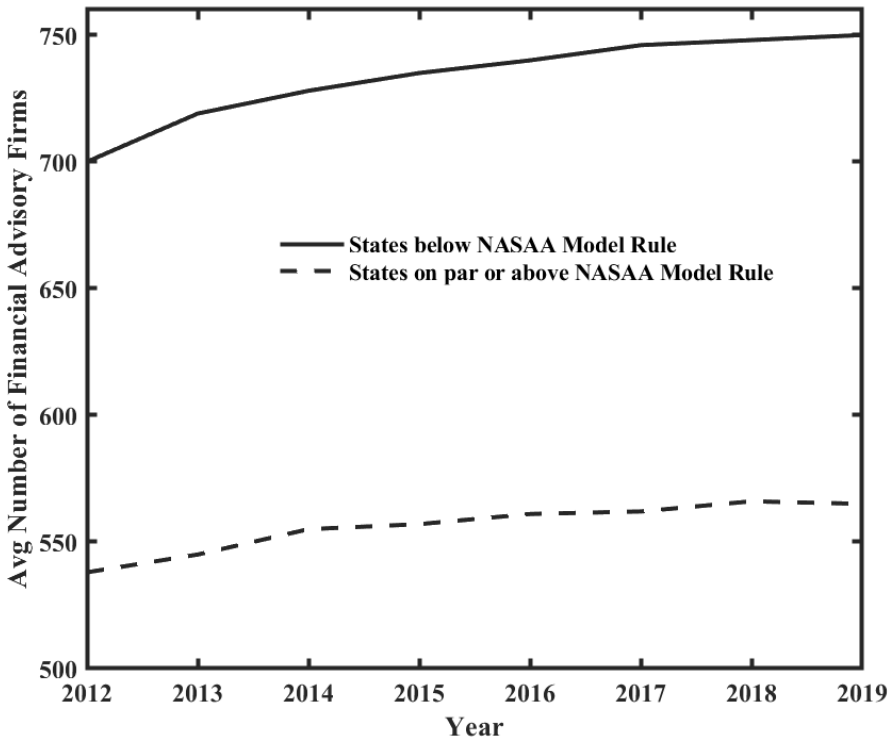
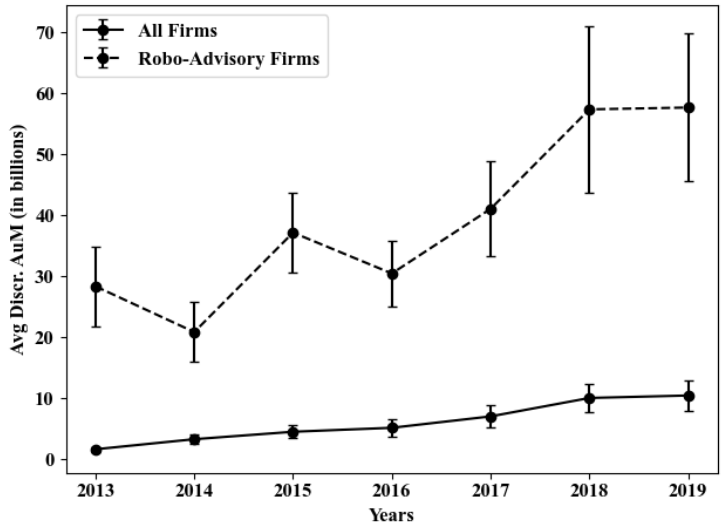
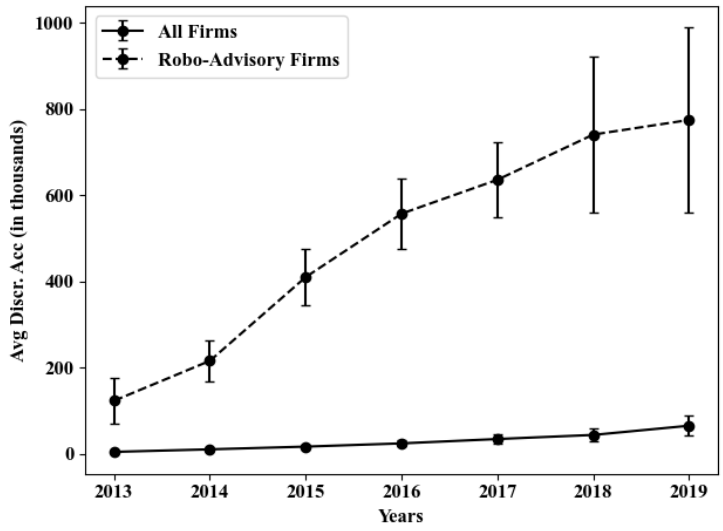


Figure 8: **Average Number of Financial Advisory Firms by Regulatory Stringency.** The figure plots the average number of registered financial advisory firms in lenient and stringent states from 2012 to 2019. Unlike robo-advisor entry, the total number of advisory firms evolves smoothly across both groups with no visible post-2016 divergence.



(a) Average Discretionary Assets Under Management



(b) Average Number of Discretionary Accounts

Figure 9: **Growth in Advisory Market Scale.** Panel A plots the average discretionary assets under management (AUM) and Panel B plots the average number of discretionary accounts over time. The figures compare trends for all advisory firms and for firms offering robo-advisory services. Both discretionary AUM and the number of discretionary accounts increase substantially over the sample period, with particularly strong growth among robo-advisory firms after 2016.

Table 1: Summary Statistics

	Mean	Std. Dev.	Min	P_{25}	P_{50}	P_{75}	Max
Panel A: Firm–State–Year Variables							
# Advisors	5.40	43.14	0	1	1	2	3,403
# Brokers	11.93	106.91	0	0	0	1	9,549
# Tot Emp	14.60	111.90	1	1	1	4	9,593
Advisor Experience	13.94	7.89	0	8	13	18	38
Hiring	1.38	15.42	0	0	0	1	2,025
Separation	0.61	4.59	0	0	0	1	555
Pr. of Misconduct	0.04	0.20	0	0	0	0	1
Change # Advisors	0.27	5.89	-555	0	0	0	1,153
Panel B: State–Year Variables							
# Firms	2,214.54	1,616.50	1	1,025	1,724	3,344	6,068
# Robo-Advisors	5.88	5.93	1	1	3	8	18
Change # Robo-Advisors	1.24	2.02	-3	0	1	2	16
$Lenient_s$	0.52	0.50	0	0	1	1	1
$Post2016_t$	0.42	0.49	0	0	0	1	1
$IV_{s,t}$	0.20	0.40	0	0	0	0	1
Panel C: Firm–Year Variables							
Discr. AUM (in B)	10.80	63.30	0	0.04	0.21	1.32	4530
Discr. Acc	18,387	192,961	0	32	404	2265	18,200,000
Panel C: Investor–State–Year Variables							
Pr. Robo Usage	0.01	0.12	0	0	0	0	1
Pr. Advisor Usage	0.54	0.50	0	0	1	1	1

This table reports summary statistics for the main variables used in the analysis. Panel A presents firm–state–year variables constructed from regulatory filings of registered investment advisers (RIAs). These variables capture the size and composition of advisory firms operating in each state, including the number of financial advisors, brokers, total employees, average advisor experience, hiring and separation flows, and the incidence of misconduct. Panel B reports state–year variables used in the instrumental-variable design. #Robo-Advisors measures the number of robo-advisory firms operating in a state-year. $Lenient_s$ is an indicator equal to one for states with lower minimum financial requirements for investment adviser registration relative to the NASAA model rule. $Post2016_t$ is an indicator for the post-2016 period, when a national financing shock relaxed funding constraints for young technology firms. $IV_{s,t}$ is the interaction $Lenient_s \times Post2016_t$, which generates variation in robo-advisor diffusion across states. Panel C reports firm–year measures of discretionary regulatory assets under management (AuM) and discretionary accounts. These variables capture the scale of advisory activity within firms. Panel D reports investor–state–year variables measuring the probability that investors use robo-advisory services or traditional financial advisors.

Table 2: Impact of Robo-Advisors on Financial Advisor Employment

	State-Year $\#FA_{s,t}$	Firm-State-Year $\#FA_{i,s,t}$
Panel A: OLS Estimates		
$R_{s,t}$	472.44***	0.18**
	(82.30)	(0.08)
Panel B: First Stage		
$IV_{s,t}$	0.81***	0.43***
	(0.25)	(0.01)
First-stage F-statistic	10.48	2,460
Panel C: 2SLS Estimates		
$\widehat{R}_{s,t}$	797.81**	1.13**
	(389.43)	(0.52)
State-Year Controls	Yes	Yes
Firm Fixed Effects	No	Yes
State Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes
Firm × Year Fixed Effects	No	Yes
Observations	408	305,716

The dependent variable (#FA) is the number of financial advisors. $R_{s,t}$ denotes the number of robo-advisory firms operating in state s and year t. $IV_{s,t}$ is the interaction of an indicator for lenient states (below NASAA financial requirements) and an indicator for years 2016 and after. State-year controls include financial-sector employment, the number of advisory firms, personal consumption expenditure on financial services, population aged over 25, GDP, unemployment rate, share of college-educated individuals, and share of female employment. Standard errors are clustered at the state level in the state-year specification and at the firm level in the firm-state-year specification. Robust standard errors are reported in parentheses. Significance is denoted as: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 3: Robo-Advisor Diffusion and Advisor Employment: Robustness to Firm-Level Scale and Demand Controls

	(1) $\#\text{FA}_{i,s,t}$	(2) $\#\text{FA}_{i,s,t}$	(3) $\#\text{FA}_{i,s,t}$
$\widehat{R}_{s,t}$	1.01**	1.10**	1.10*
	(0.52)	(0.52)	(0.60)
$\#\text{Brkrs}_{i,s,t}$		0.24***	0.27***
		(0.06)	(0.06)
$\log(1+\text{AuM})_{i,t}$			-0.19***
			(0.04)
$\log(1+\text{Acc})_{i,t}$			0.72***
			(0.16)
Industry Controls	Yes	Yes	Yes
Demand Proxies	Yes	Yes	Yes
Economic Variables	No	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Firm × Year Fixed Effects	Yes	Yes	No
Observations	305,716	305,716	210,936

The dependent variable is the number of financial advisors (#FA) at the firm–state level. $\widehat{R}_{s,t}$ is the fitted number of robo-advisors from the first stage. #Brkrs denotes the number of brokers at the firm–state level. $\log(1 + \text{AuM})$ is the natural logarithm of one plus discretionary regulatory assets under management, and $\log(1+\text{Acc})$ is the natural logarithm of one plus discretionary regulatory accounts. All specifications include state-year industry controls (financial-sector employment and number of advisory firms), demand proxies (personal consumption expenditure on financial services and population over age 25), and state economic variables as indicated. Standard errors are clustered at the firm level. Significance is denoted as: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 4: Robo Diffusion and Expansion of Advisory Assets and Accounts

Panel A: State-Level Assets and Accounts at Robo-Advisory Firms		
	Discretionary AuM (Billions)	Discretionary Accounts (Thousands)
$R_{s,t}$	50.1***	11.90***
	(1.43)	(0.31)
State-Year Controls	Yes	Yes
State FE	Yes	Yes
Year FE	Yes	Yes
Panel B: Firm-Level Assets and Accounts		
	Discretionary AuM (Millions)	Discretionary Accounts (Thousands)
$R_{s,t}$	56.50	-0.10
	(330.00)	(0.79)
$R_{s,t} \times \mathbf{1}\{\text{Robo}\}$	864.00**	2.24**
	(364.00)	(0.87)
Firm-Year Controls	Yes	Yes
Firm-State-Year Controls	Yes	Yes
State Controls	Yes	Yes
Firm FE	Yes	Yes
State FE	Yes	Yes
Year FE	Yes	Yes

Panel A examines the relationship between robo-advisor diffusion and discretionary assets under management (AuM) and discretionary accounts at the state level. Panel B examines the same outcomes at the firm level. The interaction term identifies whether growth is concentrated among firms offering robo-advisory services. Standard errors are clustered at the state level in Panel A and at the firm level in Panel B.

Table 5: Scope of the Employment Effect: Advisors, Brokers, and Total High-Skilled Employment

	(1) Advisory-Only Firms $\#\text{FA}_{i,s,t}$	(2) Brokers $\#\text{Brkr}_{i,s,t}$	(3) Total High-Skilled $\#\text{Emp}_{i,s,t}$
$\widehat{R}_{s,t}$	4.06***	0.12	0.69
	(1.38)	(0.67)	(0.83)
$\#\text{Brkrs}_{i,s,t}$	0.25***		
	(0.06)		
State-by-Year Controls	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Firm × Year Fixed Effects	Yes	Yes	Yes
Observations	284,916	305,716	305,716

The dependent variable in Column (1) is the number of financial advisors (#FA) at firms that provide advisory services (exclusive brokerage firms excluded). Column (2) uses the number of brokers (#Brkr) as the dependent variable. Column (3) uses total high-skilled employment (#Emp), defined as the sum of financial advisors and brokers. $\widehat{R}_{s,t}$ is the fitted number of robo-advisors from the first stage. State-by-year controls include financial-sector employment, number of advisory firms, personal consumption expenditure on financial services, population over age 25, GDP, unemployment rate, share of college-educated individuals, and share of female employment. Standard errors are clustered at the firm level. Significance denoted as: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6: Robo-Advisor Diffusion and Investor Demand for Financial Advisors

	(1) 2SLS $\mathbf{1}\{Use\ FA = 1\}_{j,s,t}$	(2) OLS $\mathbf{1}\{Use\ FA = 1\}_{j,s,t}$
$R_{s,t}$	0.35***	
	(0.09)	
$\mathbf{1}\{Use\ Robo = 1\}_{j,s,t}$		0.06***
		(0.02)
State-Year Controls	Yes	Yes
Individual Characteristics	Yes	Yes
Individual-Year Characteristics	Yes	Yes
State Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes

The dependent variable equals one if individual j in state s uses financial advisory or brokerage services in year t. Column (1) reports 2SLS estimates of the effect of state-level robo-advisor diffusion ($R_{s,t}$) on the probability of using a financial advisor. Column (2) reports OLS estimates of the association between individual robo-advisor usage and financial advisor usage. $\mathbf{1}\{Use\ Robo = 1\}_{j,s,t}$ equals one if the individual reports using robo-advisory services. State-year controls include financial-sector employment, number of advisory firms, personal consumption expenditure on financial services, population over age 25, GDP, unemployment rate, share of college-educated individuals, and share of female employment. Individual controls include demographic and financial characteristics. Robust standard errors are reported in parentheses. Significance denoted as: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 7: Heterogeneous Effects: Firms Offering Services Similar to Robo-Advisory Models

	(1) Individual Clientele $\#\text{FA}_{i,s,t}$	(2) Wrap Programs $\#\text{FA}_{i,s,t}$	(3) Pension Services $\#\text{FA}_{i,s,t}$
$R_{s,t}$	1.10*	1.01*	1.00*
	(0.60)	(0.58)	(0.60)
$\mathbf{1}\{Indv = 1\} \times R_{s,t}$	1.13***		
	(0.24)		
$\mathbf{1}\{Wrap = 1\} \times R_{s,t}$		0.82***	
		(0.28)	
$\mathbf{1}\{Pension = 1\} \times R_{s,t}$			0.60**
			(0.24)
Firm-Year Controls	Yes	Yes	Yes
Firm-State-Year Controls	Yes	Yes	Yes
State-Year Controls	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes

The dependent variable is the number of financial advisors at the firm–state level. Column (1) interacts robo-advisor diffusion with an indicator for firms serving predominantly individual clientele (defined as firms with more than 50% individual clients). Column (2) interacts robo diffusion with an indicator for firms offering wrap-fee programs. Column (3) interacts robo diffusion with an indicator for firms offering pension services. $R_{s,t}$ denotes the number of robo-advisors in state s at time t. Firm-year controls include $\log(1 + \text{AuM})$ and $\log(1 + \text{Accounts})$. Firm-state-year controls include the number of brokers. State-year controls include financial-sector employment, number of advisory firms, personal consumption expenditure on financial services, population over age 25, GDP, unemployment rate, share of college-educated individuals, and share of female employment. Standard errors are clustered at the firm level. Significance denoted as: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 8: Service Orientation and Robo-Advisor Exposure: Retail vs. High-Net-Worth Firms

	(1) Retail-Oriented Firms $\#\text{FA}_{i,s,t}$	(2) HNW-Oriented Firms $\#\text{FA}_{i,s,t}$
$R_{s,t}$	1.10*	1.01*
	(0.60)	(0.58)
$\mathbf{1}\{Retail = 1\} \times R_{s,t}$	0.82***	
	(0.27)	
$\mathbf{1}\{HNW = 1\} \times R_{s,t}$		-0.18
		(0.18)
Firm-Year Controls	Yes	Yes
Firm-State-Year Controls	Yes	Yes
State-Year Controls	Yes	Yes
Firm Fixed Effects	Yes	Yes
State Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes

The dependent variable is the number of financial advisors at the firm–state level. Column (1) interacts robo-advisor diffusion with an indicator for firms serving predominantly retail clientele (defined as firms with more than 75% individual clients). Column (2) interacts robo diffusion with an indicator for firms serving predominantly high-net-worth (HNW) clientele (more than 75% HNW clients). $R_{s,t}$ denotes the number of robo-advisors in state s at time t. Firm-year controls include log(1 + AuM) and log(1 + Accounts). Firm-state-year controls include the number of brokers. State-year controls include financial-sector employment, number of advisory firms, personal consumption expenditure on financial services, population over age 25, GDP, unemployment rate, share of college-educated individuals, and share of female employment. Standard errors are clustered at the firm level. Significance denoted as: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 9: Organizational Complementarity: Robo-Adopting Firms and Advisory Employment

	(1) Robo-Adopting Firms $\#\text{FA}_{i,s,t}$
$R_{s,t}$	3.49***
	(0.37)
$\mathbf{1}\{RoboAdopt = 1\} \times R_{s,t}$	22.11**
	(10.35)
Firm-Year Controls	Yes
Firm-State-Year Controls	Yes
State-Year Controls	Yes
Firm Fixed Effects	Yes
State Fixed Effects	Yes
Year Fixed Effects	Yes

The dependent variable is the number of financial advisors at the firm–state level. The table interacts state-level robo-advisor diffusion with an indicator for firms that provide robo-advisory services. $R_{s,t}$ denotes the number of robo-advisors in state s at time t. Firm-year controls include $\log(1+\text{AuM})$ and $\log(1+\text{Accounts})$. Firm-state-year controls include the number of brokers. State-year controls include financial-sector employment, number of advisory firms, personal consumption expenditure on financial services, population over age 25, GDP, unemployment rate, share of college-educated individuals, and share of female employment. Standard errors are clustered at the firm level. Significance denoted as: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 10: Robo-Advisor Diffusion and Labor Adjustment Margins: Separations and Hiring

	(1) Separations$_{i,s,t}$	(2) Hiring$_{i,s,t}$
$R_{s,t}$	-0.20**	0.40**
	(0.09)	(0.20)
Firm-State-Year Controls	Yes	Yes
State-Year Controls	Yes	Yes
Firm Fixed Effects	Yes	Yes
State Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes
Firm × Year Fixed Effects	Yes	Yes

The dependent variable in Column (1) is the number of financial advisors separating from a firm–state between $t-1$ and t, defined as advisors present in the prior year but not in the current year. Column (2) uses the number of financial advisors hired, defined as advisors present in the current year but not in the prior year. $R_{s,t}$ denotes the number of robo-advisors in state s at time t. Firm-state-year controls include the number of advisors and brokers. State-year controls include financial-sector employment, number of advisory firms, personal consumption expenditure on financial services, population over age 25, GDP, unemployment rate, share of college-educated individuals, and share of female employment. Standard errors are clustered at the firm level. Significance denoted as: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 11: Robo-Advisor Diffusion, Advisor Experience, and Misconduct

	(1) Avg. Experience$_{i,s,t}$	(2) Misconduct$_{i,s,t}$
$R_{s,t}$	0.37***	0.00
	(0.12)	(0.00)
Firm-Year Controls	Yes	Yes
Firm-State-Year Controls	Yes	Yes
State-Year Controls	Yes	Yes
Firm Fixed Effects	Yes	Yes
State Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes

Column (1) uses the average experience of financial advisors at the firm–state level as the dependent variable. Column (2) uses an indicator equal to one if at least one misconduct complaint is recorded at the firm–state level during the year. $R_{s,t}$ denotes the number of robo-advisors in state s at time t. Firm-year controls include log(1 + AuM) and log(1 + Accounts). Firm-state-year controls include the number of advisors and brokers. State-year controls include financial-sector employment, number of advisory firms, personal consumption expenditure on financial services, population over age 25, GDP, unemployment rate, share of college-educated individuals, and share of female employment. Standard errors are clustered at the firm level. Significance denoted as: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Internet Appendix

A1 Additional Institutional Details and Industry Evolution

This section provides additional descriptive evidence on the evolution of robo-advisory services and the composition of firms in the U.S. financial advisory industry. These facts provide further context for the empirical analysis in the main text by illustrating how robo-advisory services diffused within the industry over time and how firms offering robo services differ from traditional advisory firms.

A1.1 Evolution of Robo-Advisory Firms

Robo-advisory firms first appeared in the mid-2000s, with the earliest entrants launching around 2006. Entry into the robo-advisory space accelerated substantially beginning around 2013, followed by a pronounced increase in adoption by incumbent advisory firms after 2015. Figure A1 illustrates the evolution of firms offering robo-advisory services between 2006 and 2019. The figure shows that early entry was limited, but the number of firms operating robo-advisory platforms expanded rapidly during the second half of the sample period.

A1.2 Client Composition and Service Characteristics

Firms offering robo-advisory services are disproportionately oriented toward retail clients. Figure A2 presents the distribution of firms by the share of individual clients served in 2019. Robo-advisory firms are heavily concentrated in the upper bins of the individual-client distribution, indicating that these firms primarily cater to retail investors rather than institutional clients.

The service mix of robo-advisory firms also differs from that of traditional advisory firms. Figure A3 presents the distribution of firm characteristics in 2019. Firms offering robo-advisory services are frequently large advisory firms and are highly concentrated in wrap-fee program providers, a structure commonly used for bundled portfolio management services. By contrast, pension consulting—typically associated with institutional clientele—is less prevalent among robo-advisory firms.

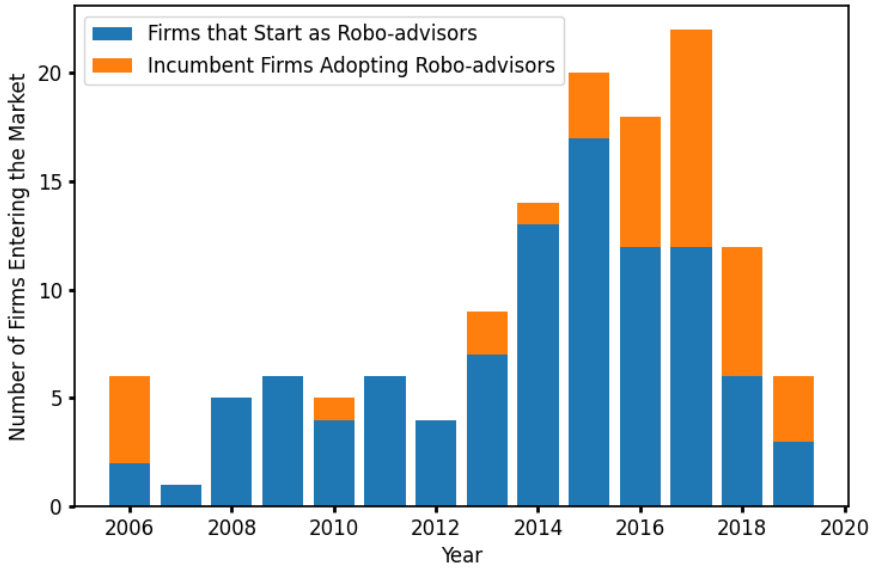


Figure A1: **Entry into Robo-Advisory Services, 2006–2019.** This figure plots the number of firms entering the robo-advisory service space each year. Robo-advisory entry is limited in the early years of the sample but increases substantially beginning around 2013, reflecting the broader expansion of fintech-based investment services.

A1.3 Firm Scale and Organizational Structure

Robo-advisory services diffuse through two distinct organizational channels: entry by fintech-oriented startups and adoption by incumbent advisory firms. Figures A4–A6 illustrate the evolution of firm scale across these categories.

Incumbent firms that adopt robo-advisory services tend to operate with substantially larger advisor bases than the average firm in the industry. By contrast, firms that start as robo-advisors typically begin with relatively few financial advisors but scale rapidly over time in terms of both assets under management and the number of client accounts. This pattern is consistent with robo-advisory technology enabling scalable retail investment services.

A1.4 Additional Industry Statistics

Table A1 provides additional evidence on the composition of advisory firms and the services they offer. Several patterns emerge. First, firms offering robo-advisory services are disproportionately

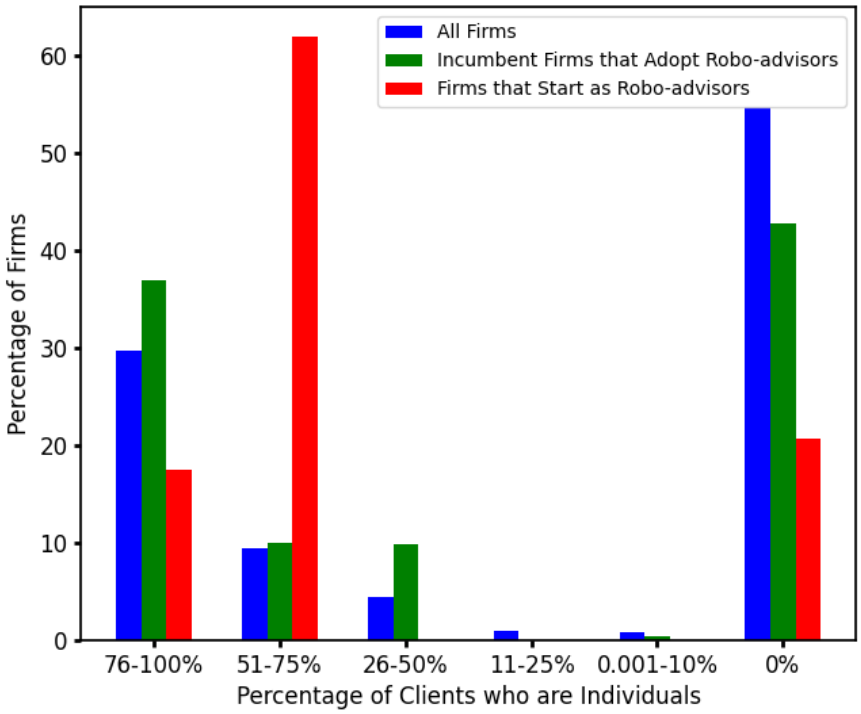


Figure A2: **Client Composition of Advisory Firms in 2019.** This figure reports the distribution of firms by the percentage of individual clients served. Firms offering robo-advisory services are disproportionately concentrated in categories with higher shares of individual clients, consistent with robo-advice being primarily targeted toward retail investors.

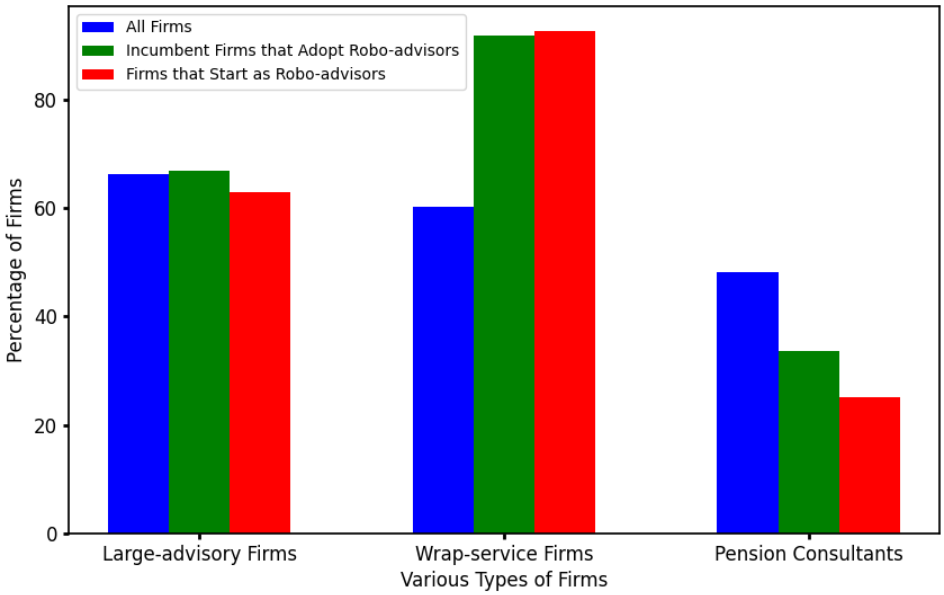


Figure A3: **Service Characteristics of Advisory Firms in 2019.** This figure shows the distribution of selected firm characteristics across advisory firms in 2019. Robo-advisory firms are strongly concentrated in wrap-fee programs and are less likely to specialize in institutional services such as pension consulting.

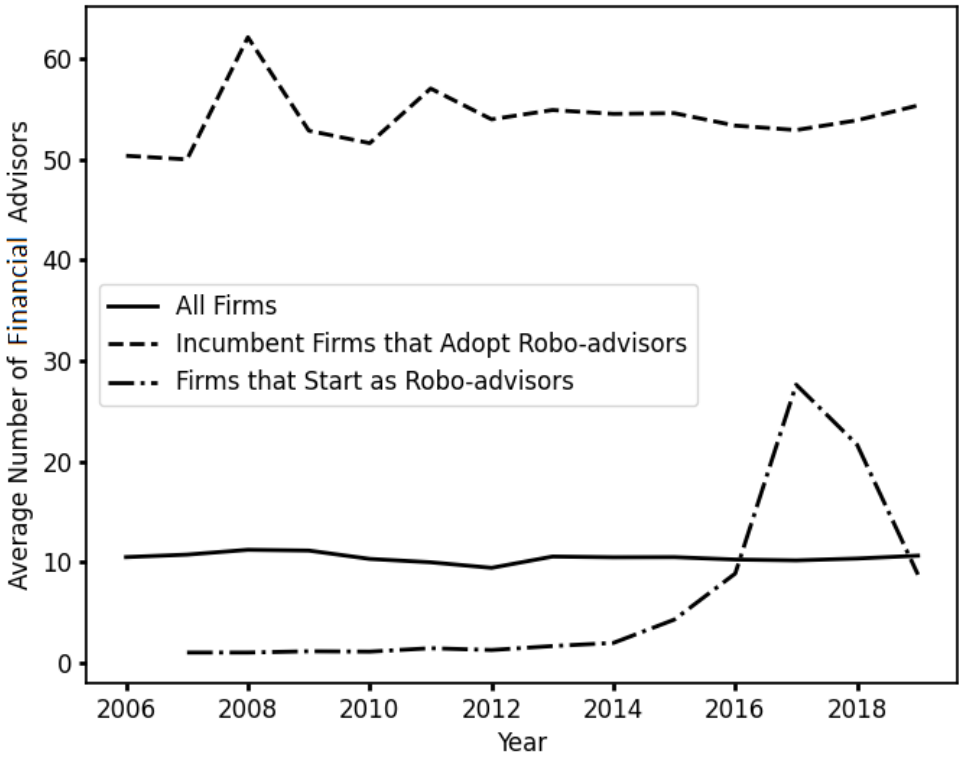


Figure A4: **Average Number of Financial Advisors by Firm Type.** This figure plots the evolution of the average number of financial advisors across firm categories between 2006 and 2019. Incumbent firms adopting robo-advisory services typically operate with larger advisor bases than robo-native entrants.

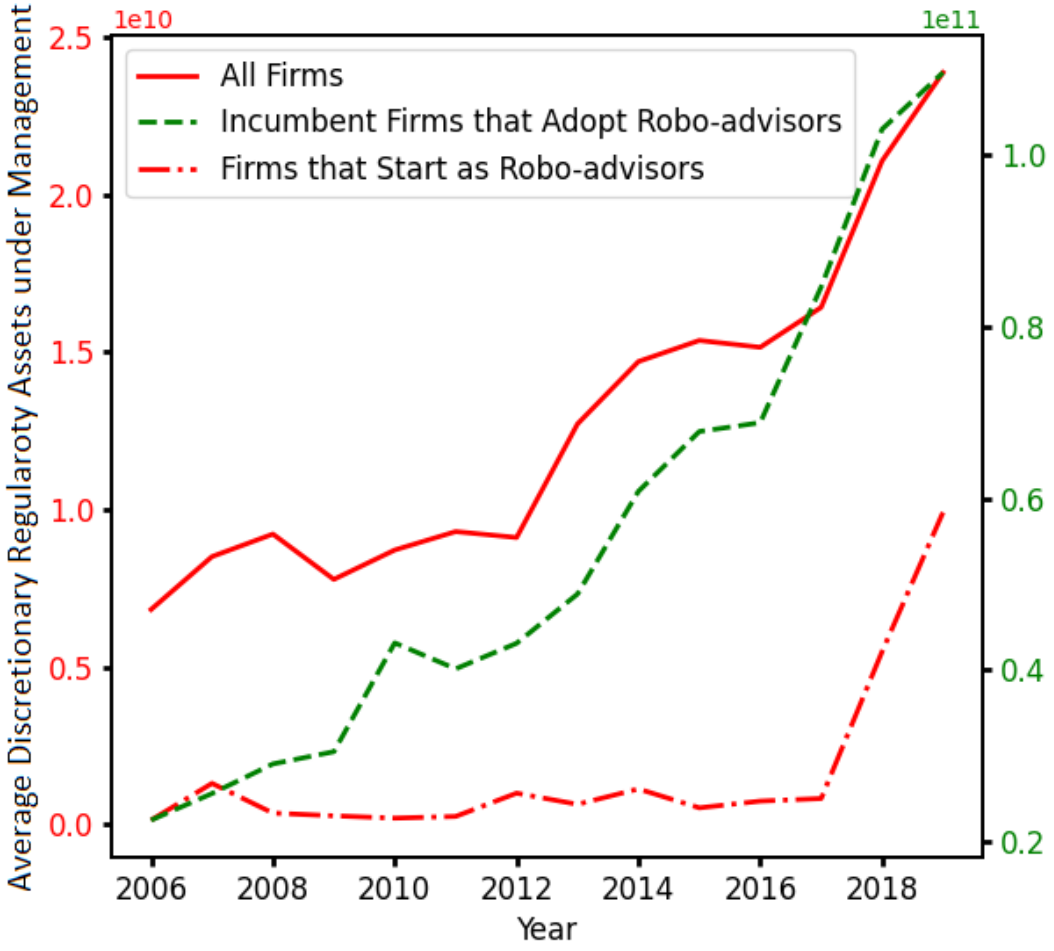


Figure A5: **Average Discretionary Regulatory Assets Under Management by Firm Type.** This figure shows the evolution of discretionary regulatory assets under management across firm categories between 2006 and 2019. Robo-native firms begin smaller but scale rapidly over time.

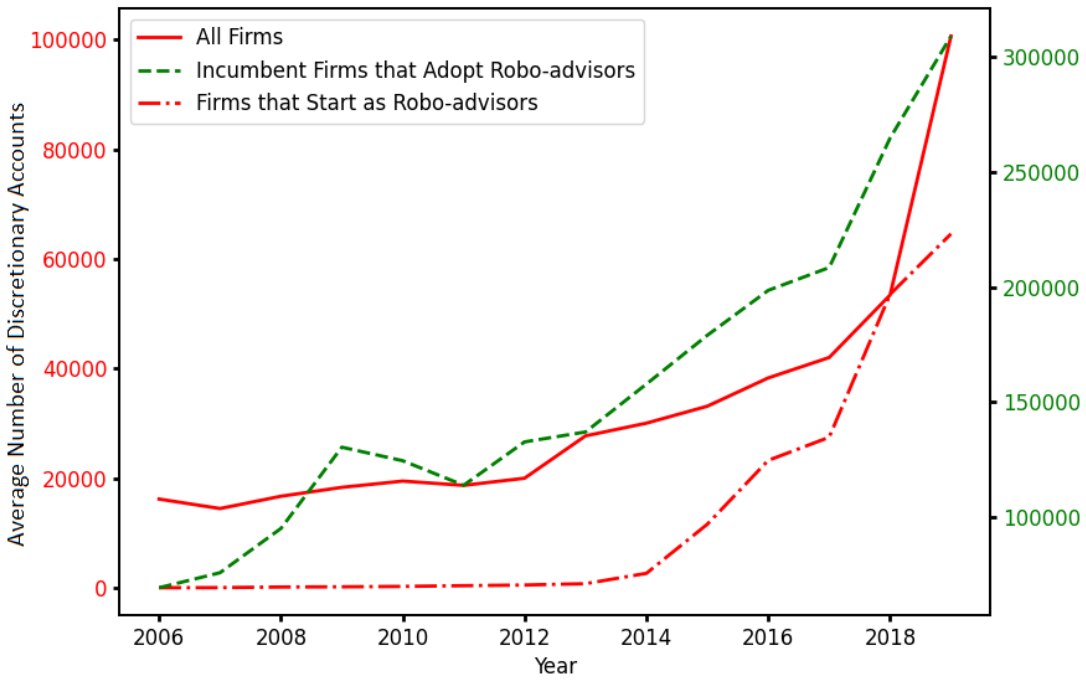


Figure A6: **Average Number of Discretionary Accounts by Firm Type.** This figure reports the average number of discretionary client accounts across firm categories between 2006 and 2019. Robo-advisory firms experience rapid growth in the number of accounts served during the later years of the sample.

oriented toward retail clients. By 2019, more than half of robo-native firms serve a client base consisting primarily of individual investors, compared with a much smaller fraction among traditional advisory firms. In contrast, high-net-worth client concentration declines substantially across the industry over the sample period.

Second, robo-advisory firms are heavily concentrated in wrap-fee programs, which bundle portfolio management and advisory services into a single fee based on assets under management. By 2019, more than 90% of firms that started as robo-advisors and nearly all hybrid adopters offer wrap-fee programs. Because wrap-fee structures are commonly used for scalable retail portfolio management, this pattern is consistent with robo-advisory services targeting standardized investment solutions for individual investors.

Third, robo-advisory firms are less likely to specialize in institutional services such as pension consulting. While roughly two-thirds of firms in the broader industry offered pension consulting services earlier in the sample, this share declines over time and remains substantially lower among robo-native firms.

Table A2 further illustrates the expansion of robo-advisory services over time by comparing industry characteristics before and after the mid-2010s expansion of robo platforms. The number of firms offering robo-advisory services increases substantially between the early and later subsamples, rising from 20 firms during 2012–2015 to 69 firms during 2016–2019. Adoption by incumbent firms also increases sharply, with the number of hybrid firms offering robo services rising from 68 to 338.

This expansion is accompanied by rapid scaling in client accounts and assets under management among robo-native firms. Discretionary regulatory assets under management at firms that began as robo-advisors increase from approximately $1 billion to $95 billion, while discretionary client accounts expand from roughly 10 thousand to more than 280 thousand, indicating that robo-advisory platforms scale quickly once adopted. These statistics illustrate that robo-advisory services initially emerged as a niche fintech entrant model but subsequently diffused more broadly through incumbent adoption and rapid scaling among robo-native firms.

Table A1: Industry Composition and Service Characteristics

	2010	2019
% of Firms with at least 50% Individual Clients	58.40	35.29
% of Firms started as Robo with at least 50% Individual Clients	33.33	53.61
% of Hybrid Firms with at least 50% Individual Clients	100.00	47.04
% of Firms with at least 50% High-Net-Worth Clients	16.61	0.04
% of Firms started as Robo with at least 50% HNW Clients	0.00	3.09
% of Hybrid Firms with at least 50% HNW Clients	0.00	0.00
% of Firms with Compensation as % of AuM	96.64	96.64
% of Robo Firms with Compensation as % of AuM	100.00	100.00
% of Firms offering Pension Services	65.84	45.17
% of Firms started as Robo offering Pension Services	100.00	8.25
% of Hybrid Firms offering Pension Services	100.00	65.14
% of Firms offering Wrap Fee Programs*	30.90	56.46
% of Firms started as Robo offering Wrap Fee Programs	100.00	90.72
% of Hybrid Firms offering Wrap Fee Programs*	89.83	98.48
% of Firms offering Only Advisory Services	24.35	31.12
% of Firms started as Robo offering Only Advisory Services	100.00	21.63
% of Hybrid Firms offering Only Advisory Services	24.35	31.12
% of Large Advisory Firms†	64.92	74.14
% of Large Advisory Firms started as Robo†	33.33	89.69
% of Large Hybrid-Advisory Firms†	100.00	60.95

*Rule 204-3(f) of the Investment Advisers Act of 1940 defines a wrap fee program as a program under which a client pays a single fee, typically expressed as a percentage of assets under management, covering both investment advisory services and transaction execution. †Large advisory firms have regulatory assets under management of at least $100 million.

Table A2: Growth of Robo-Advisory Firms

	2012–2015	2016–2019	Difference
# Firms	2154	2298	144***
# Robo Firms	20	69	49***
# Hybrid Firms	68	338	270***
# Advisors	5	6	1***
# Advisors in Robo Firms	4	52	48
# Advisors in Hybrid Firms	262	172	-90***
# Brokers	12	12	0
# Brokers in Robo Firms	0	0	0
# Brokers in Hybrid Firms	438	244	-194***
Discretionary Regulatory AuM ($ billions)	9	14	5***
AuM at Robo Firms ($ billions)	1	95	94**
AuM at Hybrid Firms ($ billions)	131	129	-2
Discretionary Accounts (thousands)	13	26	13***
Accounts at Robo Firms (thousands)	10	281	271***
Accounts at Hybrid Firms (thousands)	532	437	-95***
AuM per Account ($ millions)	31.80	21.10	-10.70***
AuM per Account at Robo Firms ($ millions)	0.46	0.32	-0.14***
AuM per Account at Hybrid Firms ($ millions)	0.22	6.06	5.85***

Table A3: Variable Definitions

Variable	Definition	Source
Panel A. Main outcome variables		
$\#FA_{i,s,t}$	Number of financial advisors employed by firm i in state s and year t.	Advisor counts are obtained using individual Form U4/BrokerCheck/IAPD records aggregated to the firm-state-year level.
$\#FA_{s,t}$	Aggregate number of financial advisors employed in state s and year t.	Aggregated $\#FA_{i,s,t}$ across the firms at state-level.
$\Delta\#FA_{i,s,t}$	Change in the number of financial advisors relative to the previous period.	Constructed from the advisor counts.

Continued on the next page

Variable	Definition	Source
$\text{Sep}_{i,s,t}$	Number of financial advisors separating from a firm-state between $t-1$ and t, defined as advisors present in the prior year but not in the current year.	Constructed from advisor-level employment histories.
$\text{Hire}_{i,s,t}$	Number of financial advisors hired into a firm-state between $t-1$ and t, defined as advisors present in the current year but not in the prior year.	Constructed from advisor-level employment histories.
$\text{AvgExp}_{i,s,t}$	Average experience of financial advisors at the firm-state-year level.	Constructed from advisor-level employment histories.
$\text{Misconduct}_{i,s,t}$	Indicator equal to one if at least one misconduct complaint is recorded at the firm-state-year level during year t.	FINRA BrokerCheck/IAPD.
$\#\text{Brkr}_{i,s,t}$	Number of brokers employed by firm i in state s and year t.	U4/BrokerCheck/IAPD records aggregated to the firm-state-year level.
$\#\text{Emp}_{i,s,t}$	Total high-skill employment, defined as the sum of financial advisors and brokers at the firm-state-year level.	Constructed from #FA and #Brkr.
Panel B. Main explanatory variables and instrument		
$R_{s,t}$	Number of robo-advisory providers operating in state s and year t.	Hand-collected census of robo providers. Both dedicated robo startups and incumbent firms that adopt robo platforms contribute to state-level robo presence.
$\Delta R_{s,t}$	Change in the number of robo-advisors in a state-year relative to the previous period.	Constructed from $R_{s,t}$.

Continued on the next page

Variable	Definition	Source
$\hat{R}_{s,t}$	Fitted number of robo-advisors from the first-stage regression.	Constructed from instrumented first stage and used in the second-stage 2SLS specifications.
Lenient_s	Indicator equal to one if state s has minimum financial requirements for registered investment advisers below NASAA Model Rule 202(d)-1 recommendations, and zero otherwise. Time-invariant over the sample.	Constructed using state regulatory rules which relies on NASAA classification.
Post2016_t	Indicator equal to one for years 2016 and after, and zero otherwise.	Calendar year that captures the post-PATH Act financing-shock period.
$\text{IV}_{s,t}$	Instrument defined as `Lenient_s` × `Post2016_t`. It captures the differential post-2016 expansion of robo-advisors in states with lower regulatory entry costs.	Constructed from regulation and year indicators.
Panel C. Firm-level scale and business-model variables		
$\text{AuM}_{i,t}$	Discretionary regulatory assets under management reported by firm i in year t.	Form ADV Part I.
$\log(1+\text{AuM}_{i,t})$	Natural logarithm of one plus discretionary regulatory assets under management. Constructed from AuM reported in Form ADV Part I.	
$\text{Acc}_{i,t}$	Number of discretionary regulatory accounts reported by firm i in year t.	Form ADV Part I.
$\log(1+\text{Acc}_{i,t})$	Natural logarithm of one plus discretionary regulatory accounts.	Constructed from Acc reported in Form ADV Part I.

Continued on the next page

Variable	Definition	Source
AUM per Account	Ratio of discretionary assets under management to discretionary accounts.	Constructed from `AuM` and `Acc`
RoboAdopt_i	Indicator equal to one if firm i provides robo-advisory services, that is, an incumbent advisory firm adopts robo technology or otherwise operates as a robo-providing firm.	Hand-collected robo census matched to firms
RoboFirm	Indicator for firms that began as robo-advisory providers.	Hand-collected robo census.
HybridFirm	Indicator for incumbent advisory firms that later adopted robo services.	Hand-collected robo census.
Panel D. Client-segment and service-orientation indicators		
Indv_i	Indicator equal to one if firm i serves predominantly individual clientele, defined in the paper as firms with more than 50% individual clients.	Form ADV client composition.
Retail_i	Indicator equal to one if firm i is retail-oriented, defined as firms with more than 75% individual clients.	Form ADV client composition.
HNW_i	Indicator equal to one if firm i serves predominantly high-net-worth clients, defined as firms with more than 75% high-net-worth clients.	Form ADV client composition.

Continued on the next page

Variable	Definition	Source
Wrap_i	Indicator equal to one if firm i offers wrap-fee programs. A wrap-fee program charges a specified fee not directly based on transactions and bundles advisory services with execution.	Form ADV Part I service disclosures.
Pension_i	Indicator equal to one if firm i offers pension-related services.	Form ADV service disclosures.
AdvisoryOnly_i	Indicator for firms providing advisory services only.	Form ADV business lines.
Panel E. Investor-demand variables		
$1UseFA_{j,s,t} = 1$	Indicator equal to one if individual j in state s and year t reports using a financial advisor or brokerage services.	FINRA NFCS
$1UseRobo_{j,s,t} = 1$	Indicator equal to one if individual j reports using robo-advisory services. Used in the individual-level OLS demand specification.	FINRA NFCS
Individual $\text{controls}_{j,s,t}$	Demographic and financial characteristics included in the NFCS regressions.	FINRA NFCS

End of Table

A2 Bartik-like Instrument: Construction, Validity, and Results

This appendix presents an alternative identification strategy based on a Bartik-like instrument. The goal is to assess whether the positive effect of robo-advisor diffusion on financial advisor employment

is robust when using (i) a different source of cross-sectional variation and (ii) a different source of time-series variation. Because Bartik-style designs can be sensitive when implemented in levels, we estimate all specifications in *changes*.

A2.1 Empirical Specification in Changes

Let $\Delta R_{s,t}$ denote the annual change in the number of robo-advisory firms operating in state s in year t, and let $\Delta FA_{s,t}$ denote the annual change in the number of human financial advisors. We estimate the following two-stage least squares specification:

$$\begin{aligned} \text{First Stage:} \quad & \Delta R_{s,t} = \pi\, Z_{s,t} + X'_{s,t}\Gamma + \alpha_s + \alpha_t + u_{s,t}, \\ \text{Second Stage:} \quad & \Delta FA_{s,t} = \beta\, \widehat{\Delta R}_{s,t} + X'_{s,t}\Lambda + \alpha_s + \alpha_t + \varepsilon_{s,t}, \end{aligned} \tag{6}$$

where $X_{s,t}$ includes the same set of time-varying state controls used in the main specifications (financial-industry conditions, proxies for demand for financial services, and state economic conditions), and α_s and α_t denote state and year fixed effects. Standard errors are clustered at the state level in state-year specifications. In firm–state–year specifications, the dependent variable is the change in financial advisors at the firm–state level, and we use the same control structure as in the main firm–state analysis.

A2.2 Construction of the Bartik-like Instrument

We construct $Z_{s,t}$ as the interaction of (i) a predetermined state "exposure" component and (ii) a time-varying "shock" component, following the logic of Bartik-style instruments. The instrument takes the form:

$$Z_{s,t} \equiv I_{s,2003} \times G_t^{UK}, \tag{7}$$

where $I_{s,2003}$ is the number of advisory firms in state s that reported providing advisory services through the internet in 2003, and G_t^{UK} is the growth in FinTech deal activity in the United Kingdom in year t.

Exposure component ($I_{s,2003}$). Beginning in mid-2002, a revision to Form ADV required advisers offering services through the internet to report that status. We use 2003 as the baseline year to mitigate undercounting in 2002 due to partial-year reporting. The identifying idea is that states with higher prevalence of internet-delivered advisory services in 2003 plausibly had greater latent demand and/or technological readiness for digital advice, making them more exposed to later robo diffusion.

Shock component (G_t^{UK}). A key challenge is that U.S. fintech growth may be correlated with U.S. economic conditions that could independently affect advisor employment. To reduce concerns that the time-series component reflects domestic confounds, we proxy the time-varying fintech shock using growth in FinTech deal activity in the UK. The maintained assumption is that global fintech investment cycles generate common variation in fintech growth across developed markets, while UK fintech growth is less likely to be mechanically linked to within-U.S. shocks affecting advisor employment.

A common concern in Bartik-style designs is that the exposure component (here, $I_{s,2003}$) may be correlated with long-run differences across states. Estimating in changes mitigates this concern by absorbing persistent differences with state fixed effects and focusing identification on differential responses over time driven by the interaction with the time-varying shock.

A2.3 Bartik-like Results: First Stage and Second Stage

Table A4 reports the first-stage and second-stage results estimated over the 2006–2019 period. Panel A shows that the Bartik-like instrument strongly predicts changes in robo-advisor entry, with first-stage F-statistics above conventional thresholds. Panel B shows that increases in robo-advisor entry are associated with economically and statistically significant increases in financial advisor employment in changes, both at the state-year level and at the firm–state–year level. These results are consistent with the main IV estimates and indicate that the positive employment effect is not specific to the primary instrument or the post-2012 estimation window.

Table A4: Bartik-like IV Estimates, 2006–2019

Panel A: First Stage		
	State-year $\Delta R_{s,t}$	Firm–state–year $\Delta R_{s,t}$
$Z_{s,t}$	0.05***	0.03***
	(0.01)	(0.01)
F-statistic	14.59	17.48
Panel B: Second Stage		
	$\Delta FA_{s,t}$	$\Delta FA_{i,s,t}$
$\widehat{\Delta R}_{s,t}$	483.20***	3.94***
	(146.03)	(0.81)
State-by-time controls	Yes	Yes
Firm FE		Yes
State FE	Yes	Yes
Year FE	Yes	Yes
Firm×Year FE		Yes
Observations	2,443	392,222

$Z_{s,t} = I_{s,2003} \times G_t^{UK}$ is the Bartik-like instrument, where $I_{s,2003}$ is the number of "internet advisory" firms in 2003 and G_t^{UK} is UK FinTech deal growth. State-by-time controls mirror those in the main specifications and include financial-industry conditions (e.g., total financial-sector employment and the number of advisory firms), proxies for demand for financial services (e.g., personal consumption expenditure on financial services and population aged over 25), and state economic conditions (e.g., GDP, unemployment, education, and female employment share). Standard errors are clustered at the state level (state-year) and at the firm level (firm–state–year).

A2.4 Validity and Diagnostics

The primary threat to identification in Bartik-like instruments is that the exposure component may be correlated with omitted factors that predict changes in the outcome. We report three diagnostics commonly used in this context.

Exposure correlates vs. instrument variation. We first examine whether baseline (2003) state characteristics explain substantial variation in the instrument itself. Table A5 compares regressions where the dependent variable is (i) the number of internet advisory firms in 2003 and (ii) the Bartik-like instrument. Baseline characteristics explain a sizable fraction of the variation in $I_{s,2003}$ but explain much less of the variation in $Z_{s,t}$, suggesting that the interaction meaningfully shifts variation toward the time-varying component.

Table A5: Correlation Between Baseline Characteristics and the Bartik-like Instrument

	# Internet Firms	$Z_{s,t}$	# Internet Firms	$Z_{s,t}$
Share college educated	0.40**	0.07***	0.39**	0.08***
	(0.17)	(0.02)	(0.18)	(0.02)
Share female employed	-2.80***	-0.45***	-3.33***	-0.56***
	(0.74)	(0.08)	(0.94)	(0.10)
Share employed	2.77***	0.44***	3.43***	0.60***
	(0.82)	(0.09)	(1.10)	(0.12)
GDP per capita	-45.85	-10.25*	-38.98	-8.14
	(48.62)	(5.25)	(53.55)	(5.54)
Share employed in financial services			0.82	0.07
			(2.05)	(0.21)
Consumption of financial services per capita			-1052.73	-261.99
			(2282.96)	(240.83)
Share of population age ≥ 25			42.03	9.07*
			(43.78)	(4.66)
R^2	0.38	0.09	0.39	0.09

Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Adding baseline-characteristics-by-time interactions. Next, following the logic in Goldsmith-Pinkham et al. (2020), we assess whether the estimated second-stage effect is sensitive to controlling for baseline correlates interacted with year fixed effects. If the instrument is picking up differential trends related to baseline characteristics, the coefficient on $\widehat{\Delta R}_{s,t}$ should change materially after

adding these interactions. Table A6 shows that the coefficient remains stable and statistically indistinguishable across specifications.

Table A6: 2SLS Estimates With and Without Baseline-Characteristics × Year Controls

	Controls excluded	Controls included	Difference
$\widehat{\Delta R}_{s,t}$	1.27***	1.24***	0.03
	(0.34)	(0.37)	

Robust standard errors clustered at the state level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Alternative exposure base years. Finally, we verify that results are not sensitive to the particular choice of exposure year by reconstructing the instrument using internet advisory firm counts from alternative base years (2004 and 2005). Table A7 shows that coefficient estimates are similar across base-year definitions, and differences relative to 2003 are statistically small.

Table A7: 2SLS Estimates Using Alternative Exposure Base Years

	2003 $\Delta FA_{i,s,t}$	2004 $\Delta FA_{i,s,t}$	2005 $\Delta FA_{i,s,t}$
$\widehat{\Delta R}_{s,t}$	1.25***	1.34***	1.42***
	(0.29)	(0.30)	(0.32)
Difference vs. 2003		0.09	0.08

Robust standard errors clustered at the state level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

A3 Additional Evidence on Investor Demographics and Robo Adoption

This appendix section provides additional descriptive and regression evidence on the demographic composition of investors using robo-advisors versus traditional financial advisors. These patterns support the interpretation that robo platforms primarily expand access to advice among younger and more employment-active households, consistent with the market-expansion mechanism emphasized in the main text.

A3.1 Demographic Composition of Robo vs. Advisor Users

Table A8 compares the demographic distribution of investors who report using financial advisors versus robo-advisors. Robo users are substantially younger: investors age 39 and under represent 43.5% of robo users (17.7% + 25.84%) compared to 16.1% of advisor users (5.75% + 10.36%). Conversely, older households are much less represented among robo users: investors age 65+ account for 14.6% of robo users compared to 41.1% of advisor users. Robo users are also more likely to be employed full time (57.3% vs. 34.9%) and less likely to be retired (15.8% vs. 42.3%). Education and income distributions are broadly similar across the two groups, though robo users are somewhat more diverse in race composition.

Table A8: Demographic Differences in Users of Financial Advisors vs. Robo-Advisors

	Financial Advisors	Robo-advisors
Age (≤ 29)	5.75%	17.7%
Age (30–39)	10.36%	25.84%
Age (40–49)	10.69%	17.11%
Age (50–64)	32.12%	24.76%
Age (≥ 65)	41.07%	14.59%
White Respondents	85.95%	72.61%
Black Respondents	5.64%	11.48%
High-school educated	14.76%	13.40%
College educated	84.81%	86.36%
Full-time employed	34.89%	57.30%
Retired	42.32%	15.79%
Income ($< \$50{,}000$)	22.14%	23.09%
Income ($\$50{,}000 \leq \cdot < \$150{,}000$)	63.53%	65.07%
Income ($\geq \$150{,}000$)	14.32%	11.84%

A3.2 Demographic Correlates of Robo Adoption

Table A9 reports regressions relating individual demographics to the probability of using a robo-advisor (Column 1) and to the probability of using a traditional financial advisor (Column 2), controlling for state and year fixed effects and state-year covariates. The estimates indicate that robo usage is more prevalent among younger, non-married, and employed individuals, whereas advisor usage is more prevalent among older, higher-income, more educated individuals. These patterns

align with the descriptive evidence in Table A8 and suggest that robo platforms expand participation among demographic segments that are relatively underrepresented in traditional advisor relationships.

Table A9: Demographic Correlates of Robo-Advisor Usage and Advisor Usage

	$\mathbf{1}\{\text{Robo} = 1\}$	$\mathbf{1}\{\text{Advisor} = 1\}$
Age	-0.005***	0.006***
	(0.000)	(0.001)
Female	-0.005***	0.015***
	(0.002)	(0.006)
Education	0.000	0.045***
	(0.001)	(0.003)
Married	-0.003***	-0.010***
	(0.001)	(0.003)
Income	-0.001	0.047***
	(0.001)	(0.002)
Employment	0.003***	-0.001
	(0.000)	(0.001)
State-year controls	Yes	Yes
State FE	Yes	Yes
Year FE	Yes	Yes
Observations	408	185,682

The dependent variables are indicators for robo-advisor usage and traditional financial advisor usage, respectively. State-year controls include measures of financial-industry conditions, proxies for demand for financial services, and state economic conditions (as in the main specifications). Standard errors are clustered at the state level.